



CSX Portfolio Optimization using Markowitz Mean-Variance Model

(Constructing Optimal Equity Portfolios in Emerging Markets)

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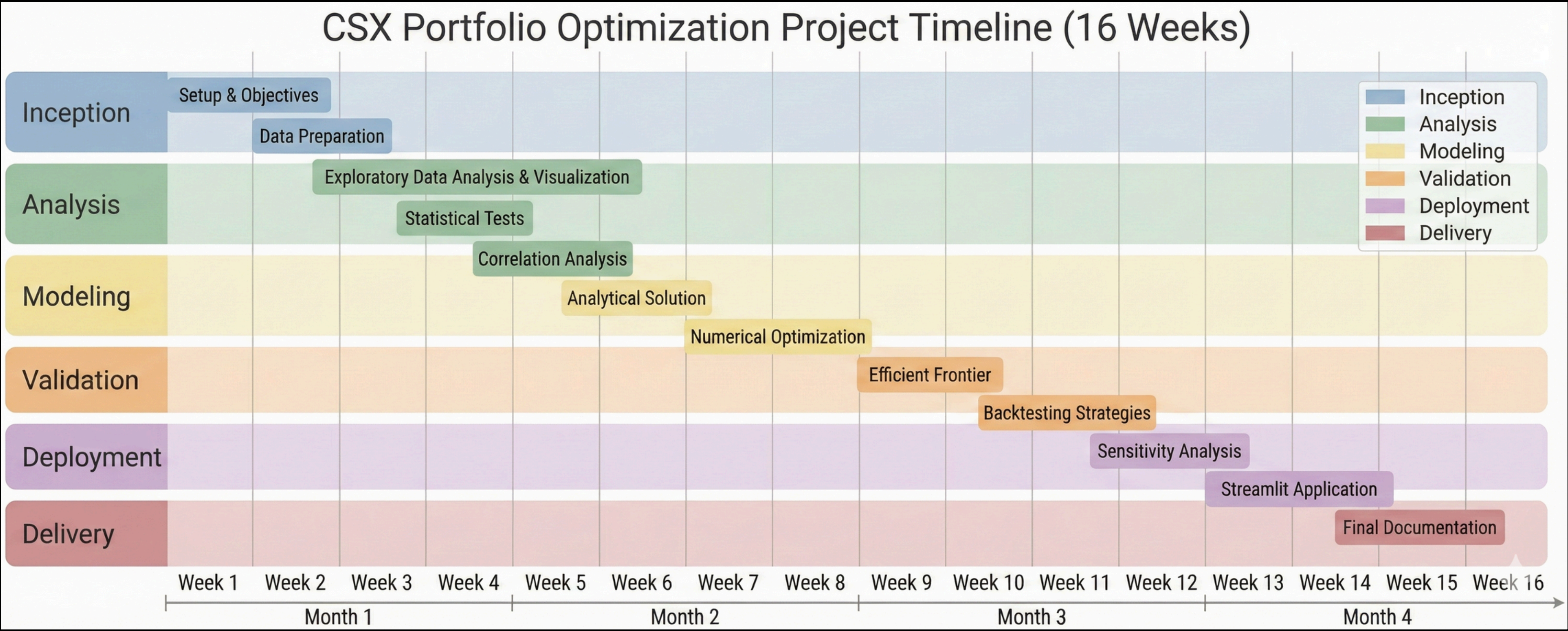
SOU Pichchomrong

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- **1. Introduction & Context**
- **2. Theoretical Foundation**
- **3. Data & Methodology**
- **4. Implementation & Results**
- **5. Sensitivity Analysis**
- **6. Conclusions & Recommendations**
- **7. Appendix & Q&A**

1.Introduction & Context

1.1 Project Timeline



1.Introduction & Context

1.2 The Cambodia Securities Exchange (CSX)



Market Context:

- Small but growing market
- Limited number of listed assets
- Unique market risks

Why This Matters ?

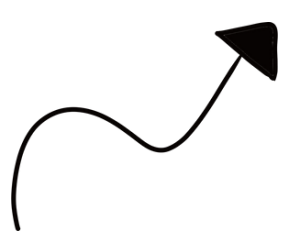
- *CSX provides a constrained environment to test portfolio theory*



1.Introduction & Context

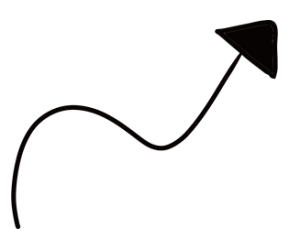
1.3 Research Motivation

Question: *"How can investors optimize portfolios in small markets?"*



The Challenge

- Limited investment opportunities
- Higher market-specific risks



Our Approach

- Apply Markowitz mean–variance portfolio theory
- Use CSX equity data to construct optimal portfolios

1.Introduction & Context

1.4 Project Objective

Primary Objective

★ **Construct an optimal portfolio using 9 CSX equities**

Specific Goals:

- **Apply the Markowitz mean–variance model.**
- **Compare analytical and numerical optimization approaches**
- **Analyze short-selling versus long-only constraints**
- **Validate results using out-of-sample testing**

1.Introduction & Context

1.5 Asset Universe

The portfolio consists of **9 equities** listed on the Cambodia Securities I



- **Data Period: June 2023 – December 2025**
- **Split: In-Sample (Training) | Out-of-Sample (Testing)**

Ticker	Company Name
ABC	ACLEDA Bank Plc.
CGSM	Canadia General Shareholding and Management
GTI	Grand Twins International
MJQE	Mekong Jewelry
PAS	Phnom Penh Autonomous Port
PEPC	Phnom Penh Electric Power Corporation
PPAP	Phnom Penh Autonomous Port
PPSP	Phnom Penh Special Economic Zone
PWSA	Phnom Penh Water Supply Authority

The dataset is split into two distinct periods for rigorous out-of-sample validation:

- **In-Sample Period (Training):** June 2023 - April 2025 (~70%)
- **Out-of-Sample Period (Testing):** May 2025 - December 2025 (~30%)

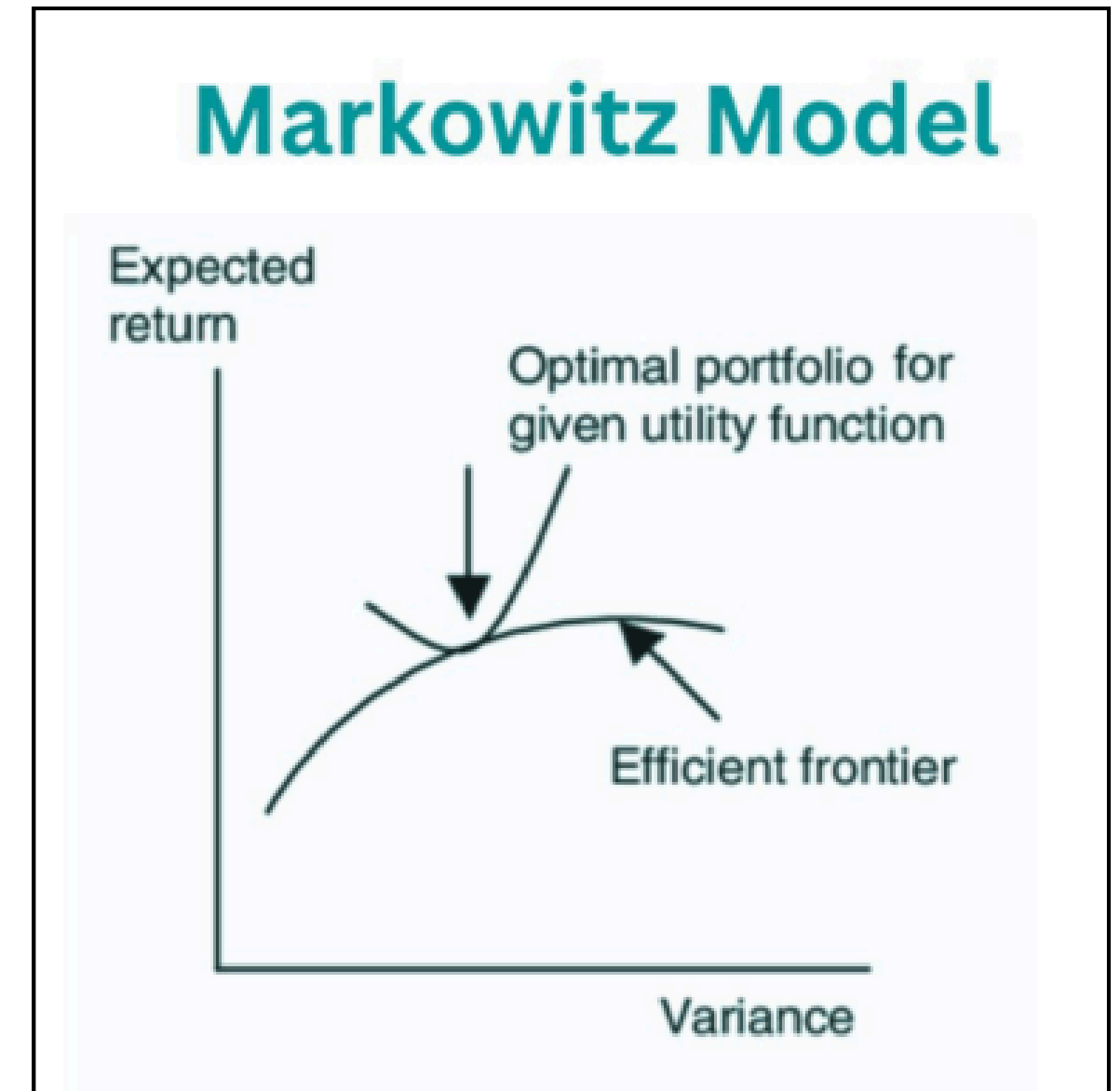
2.Theoretical Foundation

2.1 Understanding Modern Portfolio Theory

The **Markowitz Mean-Variance** Model is a foundational framework in modern portfolio theory that seeks to construct portfolios that maximize expected return for a given level of risk, or equivalently, minimize risk for a given level of expected return.

Key Principles:

- Investors are risk-averse and prefer higher returns with lower risk
- Risk is measured by portfolio variance (or standard deviation)
- Diversification can reduce portfolio risk below the weighted average of individual asset risks



“Diversification is the only free lunch in investing.” – Harry Markowitz

2.Theoretical Foundation

2.2 The Mean-Variance Framework

The expected return $\mu_V = E(K_V)$ and variance $\sigma_V^2 = \text{Var}(K_V)$ of a portfolio with weights $\mathbf{w}$ are given by

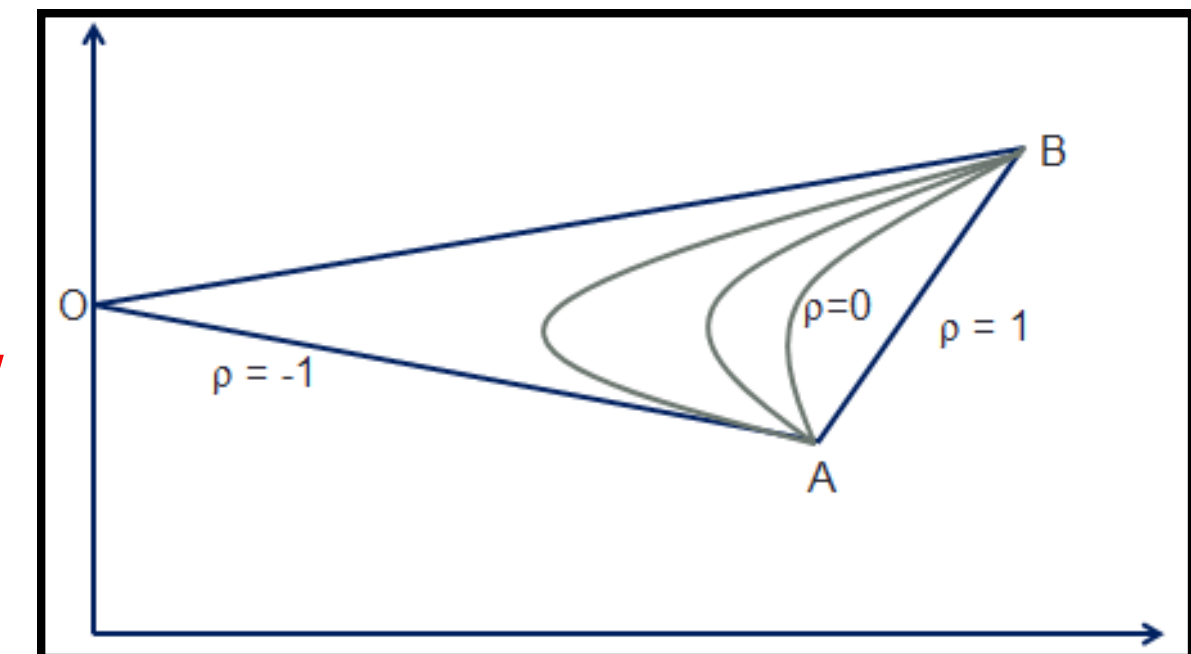
$$\mu_V = \mathbf{m}\mathbf{w}^T,$$
$$\sigma_V^2 = \mathbf{w}\mathbf{C}\mathbf{w}^T.$$

where

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} \quad \mathbf{m} = \begin{bmatrix} \mu_1 & \mu_2 & \cdots & \mu_n \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_1 & w_2 & \cdots & w_n \end{bmatrix} \quad 1 = \mathbf{u}\mathbf{w}^T,$$
$$\mathbf{u} = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}$$
$$c_{ij} = \text{Cov}(K_i, K_j) \quad \rho_{ij} = \frac{\text{Cov}(K_i, K_j)}{\sigma_i \sigma_j}$$

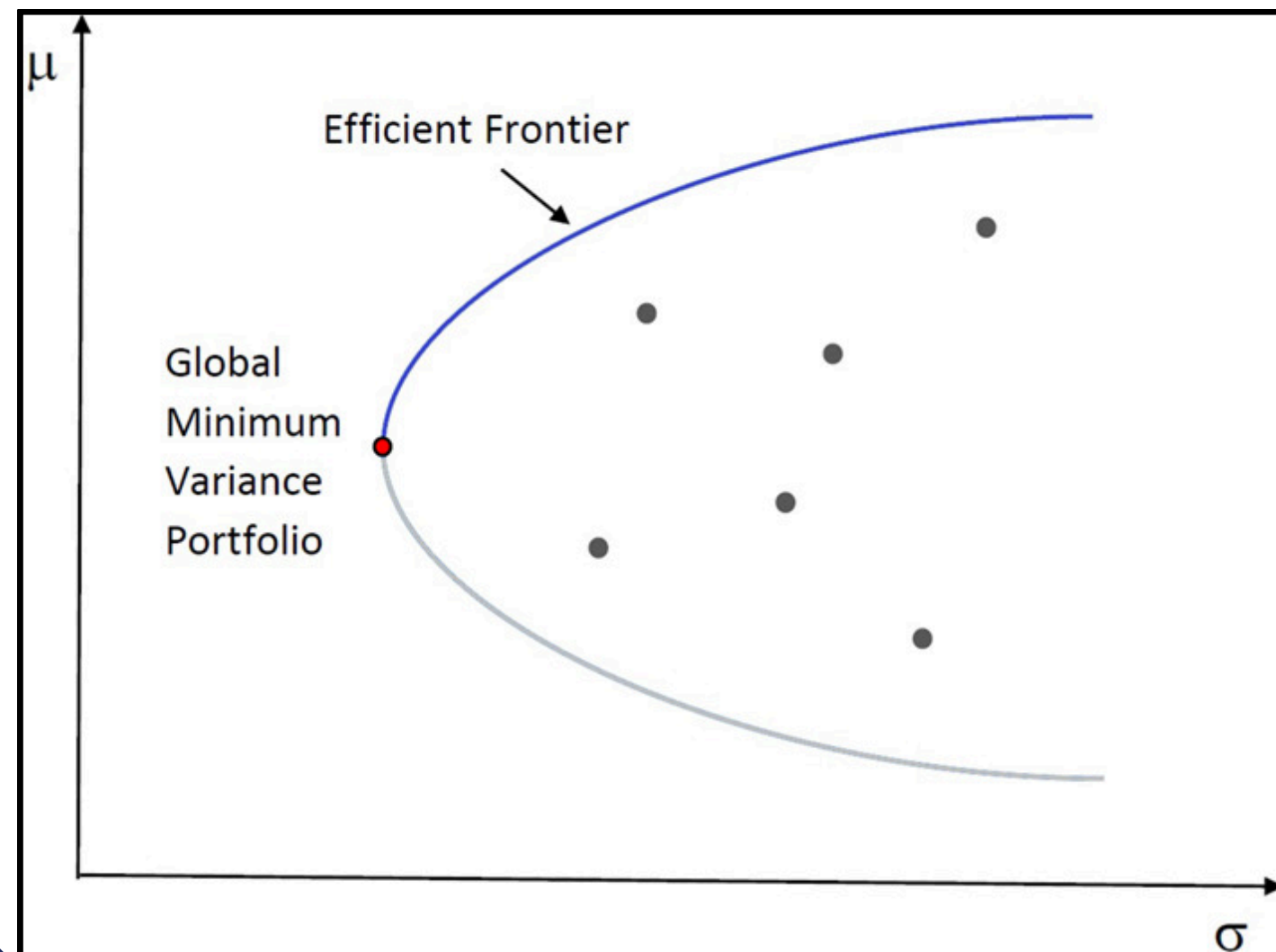
Diversification - The Only Free Lunch !

"Lower Correlation -> Greater Curvature -> Lower Portfolio Risk."



2.Theoretical Foundation

2.3 Minimum Variance Portfolio vs Minimum Variance Line



Minimum Variance Portfolio

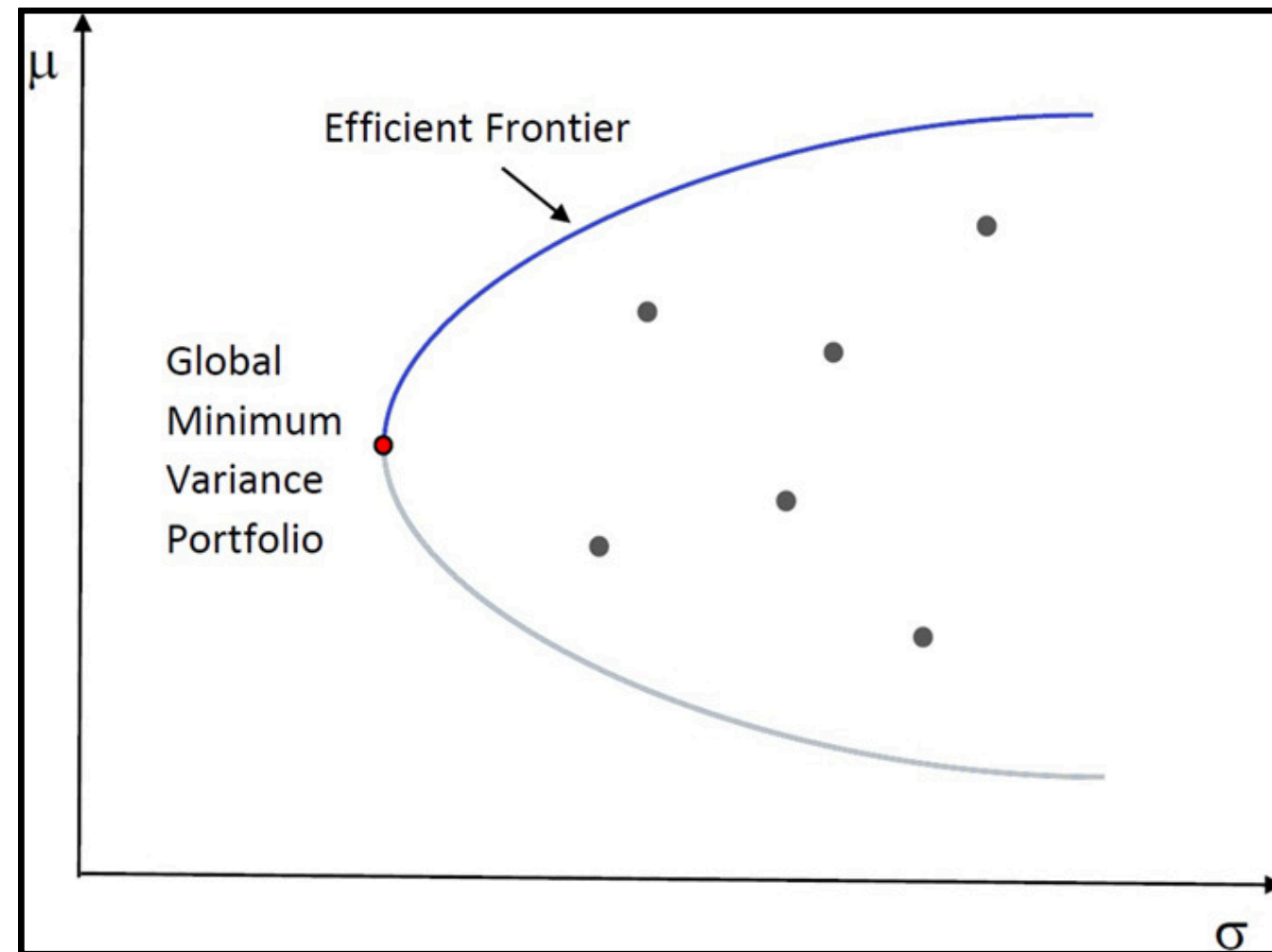
- a portfolio with the smallest variance in the attainable set
- no expected return constraint

Minimum Variance Line

- a portfolio with the smallest variance among all portfolios in the attainable set whose expected return is equal to a given number
- with expected return constraint

2.Theoretical Foundation

2.3 Minimum Variance Portfolio vs Minimum Variance Line



1. The Efficient Frontier

- The **Efficient Frontier** is the upper portion of the Minimum Variance Line, starting from the **Global Minimum Variance Portfolio**.
- All portfolios that lie on the Minimum Variance are Efficient Portfolio. They have no other **dominant portfolio**
- The lower portion of the curve is inefficient because, for any point on it, there is a point directly above it with the same risk but higher return.

2. Investor Preferences (Rational Choice)

- All rational investors choose a portfolio located on the Efficient Frontier.
- Risk-Averse: Will select portfolios on the bottom-left (near the Global Minimum Variance Portfolio) to minimize volatility.
- Risk-Tolerant: Will select portfolios on the upper-right, accepting higher volatility in exchange for higher expected returns.

$$\mu_1 \geq \mu_2 \quad \text{and} \quad \sigma_1 \leq \sigma_2.$$

2.Theoretical Foundation

2.4 Mathematical Formulation

Minimum Variance Portfolio

Optimization problem:

$$\textit{Minimize}(\textit{Variance}) = \min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$$

Subject to:

$$\mathbf{1}^\top \mathbf{w} = 1$$

Minimum Variance Line

Optimization problem:

$$\textit{Minimize}(\textit{Variance}) = \min_{\mathbf{w}} \mathbf{w}^\top C \mathbf{w}$$

Subject to:

$$\begin{cases} \mathbf{w}^\top \mu = \mu_V \\ \mathbf{1}^\top \mathbf{w} = 1 \end{cases}$$

2.Theoretical Foundation

2.5 Optimal Weight Analytical Solution (Closed-Form)

Minimum Variance Portfolio

$$w = \frac{uC^{-1}}{uC^{-1}u^T},$$

Where

- C is Covariance Matrix (C is invertible)
- m is The Expected Return Vector.
- u is The Unit Vector (vector of all 1s)

Minimum Variance Line

$$w = \frac{\begin{vmatrix} 1 & uC^{-1}m^T \\ \mu_V & mC^{-1}m^T \end{vmatrix} uC^{-1} + \begin{vmatrix} uC^{-1}u^T & 1 \\ mC^{-1}u^T & \mu_V \end{vmatrix} mC^{-1}}{\begin{vmatrix} uC^{-1}u^T & uC^{-1}m^T \\ mC^{-1}u^T & mC^{-1}m^T \end{vmatrix}},$$

Note:

- These 2 closed form solutions only handle **short-selling is allowed** condition
- The solutions are obtained by **Lagrange Multiplier** without no restricted short-sell constraint.
- To handle **no short-sell constraint**, **Quadratic Optimization** is used (numerical method).

2.Theoretical Foundation

2.6 Quadratic Optimization no short-sell (Numerical Method)

But, why Quadratic Optimization is used?

➤ Because our optimization's objective function (variance) is a quadratic function.

But how? how is variance a quadratic equation?

$$\sigma_V^2 = \text{Cov} \left(\sum_{i=1}^n w_i K_i, \sum_{j=1}^n w_j K_j \right) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j c_{ij}$$

➤ In matrix form (consistent with the code implementation),

$$\sigma_V^2 = \mathbf{w}^\top \mathbf{C} \mathbf{w}$$

This shows that portfolio variance is a quadratic function of the weights.

Optimization problem:

$$\text{Minimize}(\text{Variance}) = \min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$$

Subject to:

$$\begin{cases} \mathbf{w}^\top \mu = \mu_V \\ \mathbf{1}^\top \mathbf{w} = 1 \end{cases}$$

Short selling is not allowed and position sizes are limited:

$$0 \leq w_i \leq 1, \quad i = 1, \dots, n$$

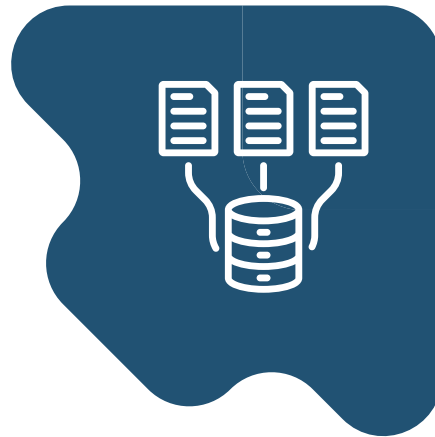
Quadratic Optimization method **SLSQP** is being used in this project.

- The **SLSQP (Sequential Least Squares Programming) algorithm** is designed to handle nonlinear objective functions together with **linear and nonlinear constraints**.

3.Data & Methodology

Data

3.1



Data Collection

Source: CSX historical price data

Time Period:

June 2023 – December 2025

Assets: 9 equities

Data Cleaning

after calculate the **lognormal return** we drop the rows if it has NaN

3.3



Outlier Treatment

use the **Winsorization** technique that reduces the impact of extreme outliers

EDA

uses visualization and summary statistics to understand datasets

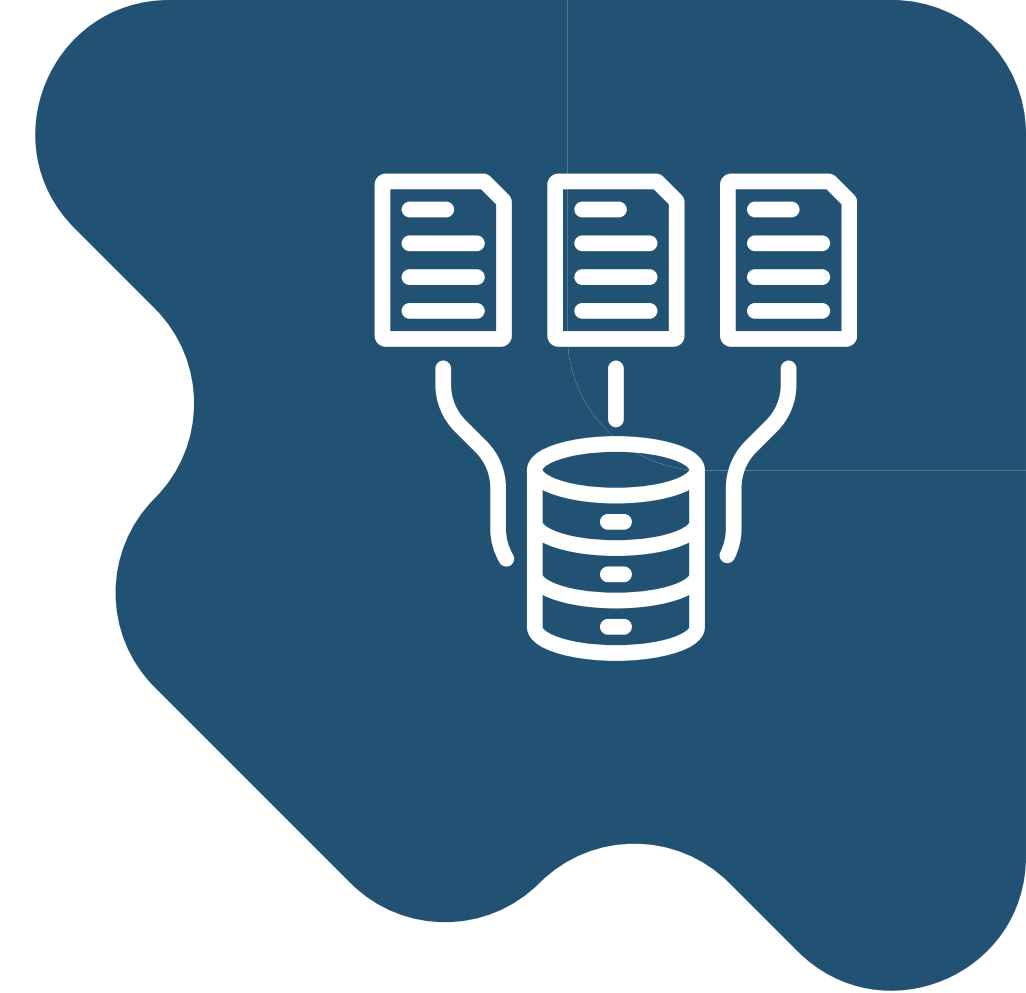
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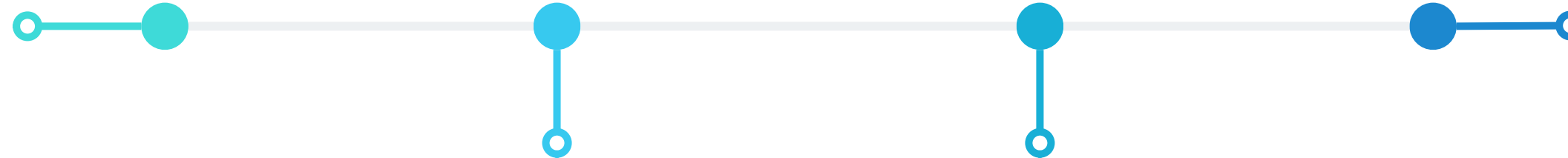
3.4



3.1 Data Collection



1/4/2021
ABC, GTI, GTI, PEPC,
PPAP, PPSP, PWSA



6/27/2023
CGSM

6/28/2023
MJQE

17-12-2025
All assets

3.2 Data Cleaning

2025-04-30

Train: (n=455)

Test: (n=153)

Train/Test Ratio: 74.8% / 25.2%

6/28/2023

All Assets

17-12-2025

All Assets

Date Range: 2023-06-28 to 2025-12-17

Shape: (608, 9)

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	ABC	607 non-null	float64
1	GTI	607 non-null	float64
2	PAS	607 non-null	float64
3	PEPC	607 non-null	float64
4	PPAP	607 non-null	float64
5	PPSP	607 non-null	float64
6	PWSA	607 non-null	float64
7	CGSM	607 non-null	float64
8	MJQE	607 non-null	float64

dtypes: float64(9)
memory usage: 47.4 KB
Missing values after cleaning: 0



Returns Calculation

Why Log Returns?

$$r_t^{(i)} = \ln\left(\frac{P_t^{(i)}}{P_{t-1}^{(i)}}\right).$$



```
1 log_returns = np.log(close_df / close_df.shift(1))
```

	ABC	GTI	PAS	PEPC	PPAP	PPSP	PWSA	CGSM	MJQE
Date									
2023-06-29	0.0000	0.0065	0.0062	0.0000	-0.0028	-0.0046	0.0000	0.0130	-0.0323
2023-06-30	0.0000	-0.0032	0.0046	-0.0137	0.0000	0.0000	0.0027	0.0296	0.0000
2023-07-03	-0.0020	0.0161	-0.0015	0.0069	-0.0070	0.0000	-0.0027	0.0042	-0.0094
2023-07-04	0.0000	-0.0064	0.0015	0.0000	-0.0014	0.0000	-0.0027	0.0083	-0.0048
2023-07-05	0.0000	-0.0032	0.0046	-0.0069	0.0070	-0.0046	0.0000	0.0244	0.0095
...	...	...	...	...	...	...	...	...	...
2025-12-11	-0.0027	-0.0057	0.0061	0.0284	0.0000	0.0000	0.0000	0.0120	0.0000
2025-12-12	0.0027	0.0085	-0.0061	-0.0035	-0.0042	0.0000	-0.0031	0.0000	0.0050
2025-12-15	-0.0250	-0.0028	-0.0155	0.0000	-0.0028	-0.0048	-0.0063	-0.0040	0.0000
2025-12-16	0.0112	-0.0028	0.0016	-0.0106	0.0014	0.0284	0.0063	-0.0080	0.0000
2025-12-17	0.0110	0.0028	0.0016	0.0000	-0.0057	0.0000	-0.0063	0.0080	0.0000

Statistical Summary of Returns

	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis	annual_return	annual_volatility
ABC	454.0	-0.000480	0.006124	-0.024151	-0.002611	0.0	0.000000	0.023110	-0.576697	5.581297	-0.120985	0.097214
CGSM	454.0	-0.000123	0.008737	-0.036203	-0.004077	0.0	0.004111	0.036636	0.222494	6.742993	-0.030964	0.138689
GTI	454.0	0.001556	0.013718	-0.051382	-0.003731	0.0	0.004381	0.054425	1.046826	4.885257	0.392147	0.217772
MJQE	454.0	-0.000123	0.005265	-0.022436	-0.004561	0.0	0.000000	0.022176	0.330144	3.949273	-0.030940	0.083572
PAS	454.0	-0.000086	0.004975	-0.016526	-0.001770	0.0	0.001754	0.016556	0.098843	2.193048	-0.021795	0.078983
PEPC	454.0	-0.000287	0.019166	-0.062322	-0.008369	0.0	0.007737	0.062613	0.460595	2.358607	-0.072449	0.304243
PPAP	454.0	-0.000328	0.008534	-0.028659	-0.004231	0.0	0.002904	0.028683	0.180430	2.407896	-0.082751	0.135476
PPSP	454.0	-0.000357	0.007639	-0.027325	-0.004587	0.0	0.004561	0.027315	0.222911	2.217518	-0.090024	0.121270
PWSA	454.0	-0.000312	0.004763	-0.016240	-0.002841	0.0	0.002747	0.015734	-0.022278	2.204876	-0.078671	0.075614

3.3 Outlier Treatment

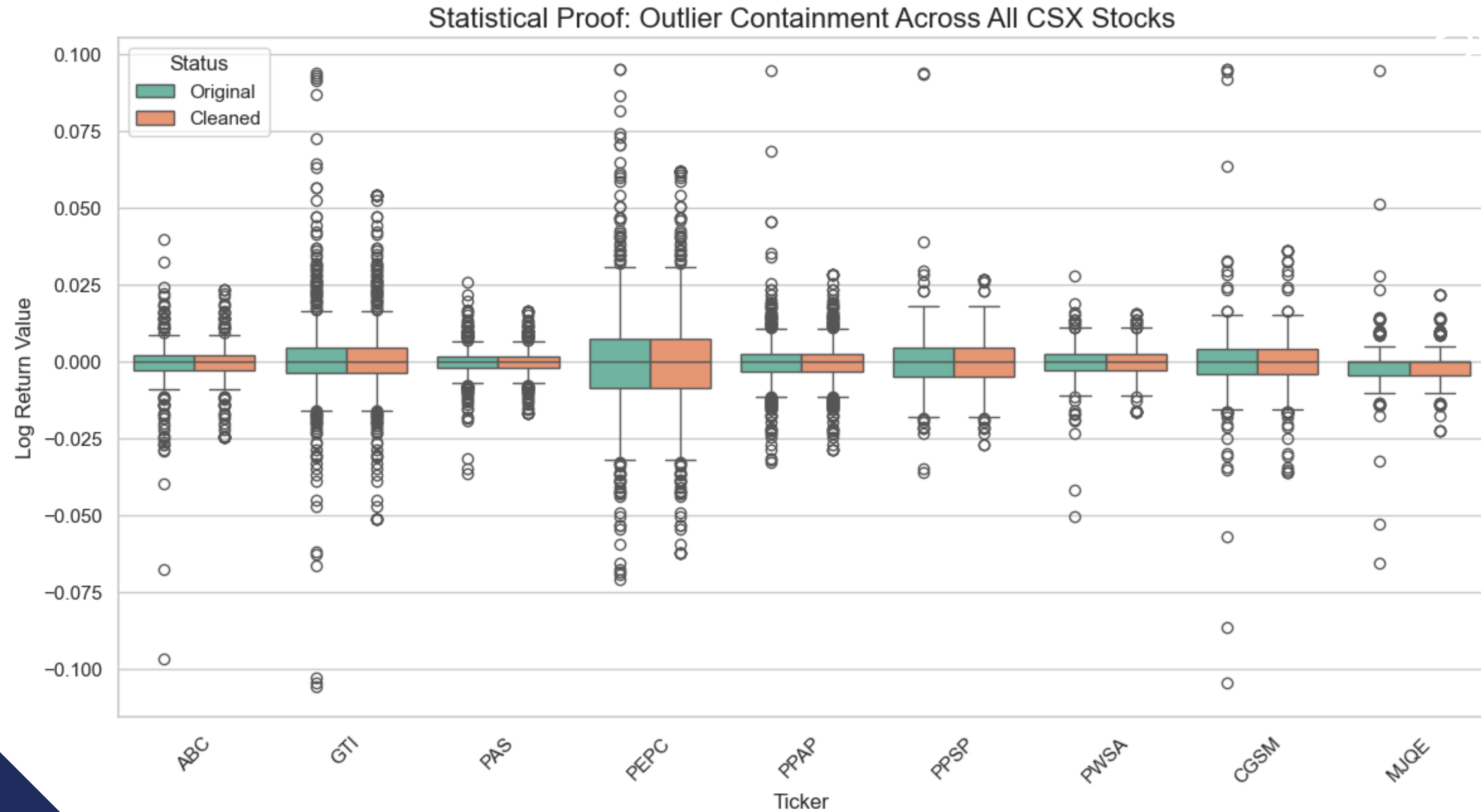


Winsorization is a statistical data transformation technique that reduces the impact of extreme outliers by pulling them at a specified 3 standard deviation rather than removing them.

✓ 3-Sigma Outlier Treatment Applied Successfully!

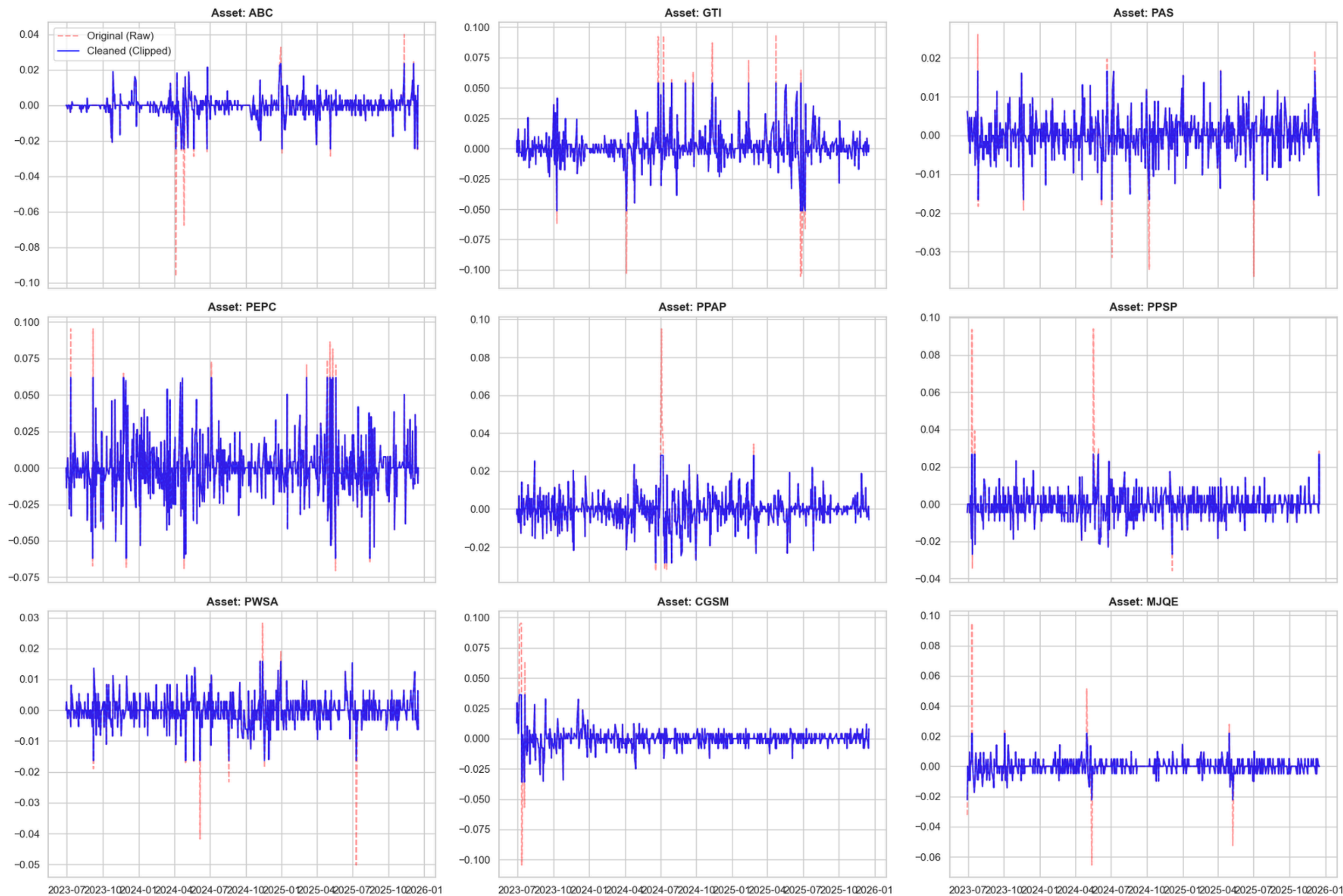
	ABC	CGSM	GTI	MJQE	PAS	PEPC	PPAP	PPSP	PWSA
count	607.000000	607.000000	607.000000	607.000000	607.000000	607.000000	607.000000	607.000000	607.000000
mean	-0.000341	-0.000030	0.001379	-0.000190	0.000076	-0.000120	-0.000242	-0.000243	-0.000166
std	0.006122	0.007803	0.014436	0.005022	0.004956	0.019631	0.007912	0.007081	0.004548
min	-0.024151	-0.036203	-0.051382	-0.022436	-0.016526	-0.062322	-0.028659	-0.027325	-0.016240
25%	-0.002660	-0.003984	-0.003584	-0.004566	-0.001729	-0.008264	-0.002946	-0.004587	-0.002841
50%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.002068	0.004098	0.004570	0.000000	0.001765	0.007605	0.002712	0.004535	0.002755
max	0.023110	0.036636	0.054425	0.022176	0.016556	0.062613	0.028683	0.027315	0.015734

BOXPLOT COMPARISON



TIME-SERIES

Time-Series Proof: Signal Preservation vs. Spike Capping



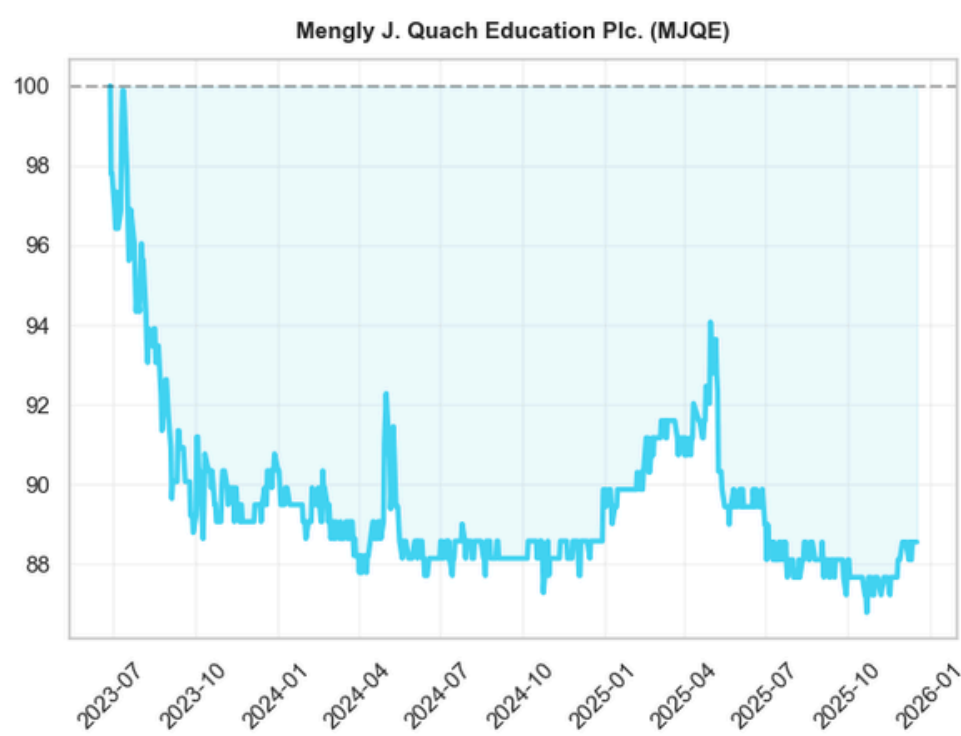
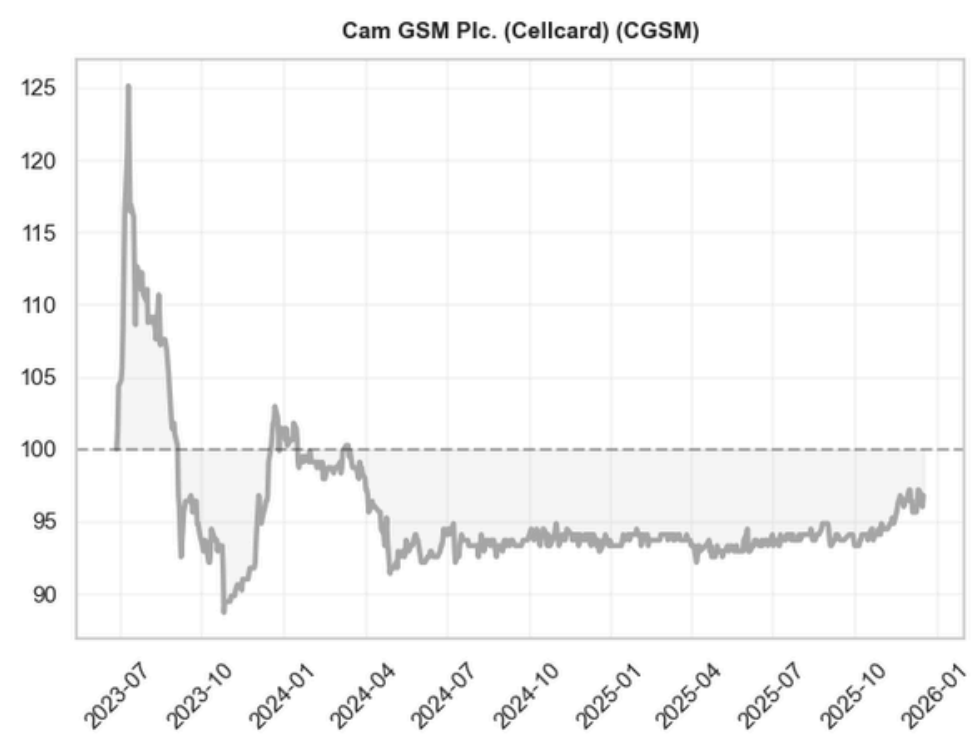
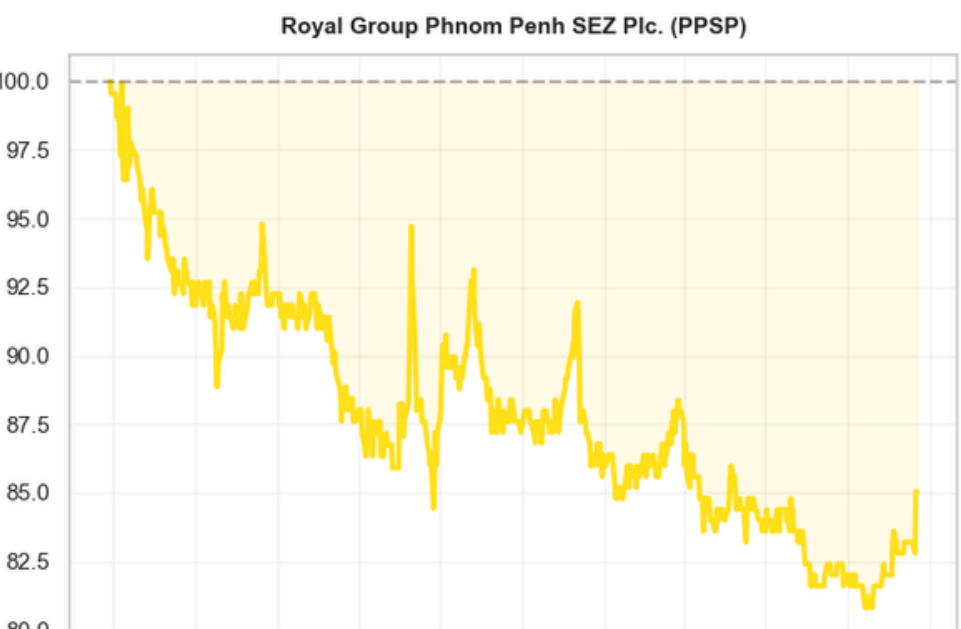
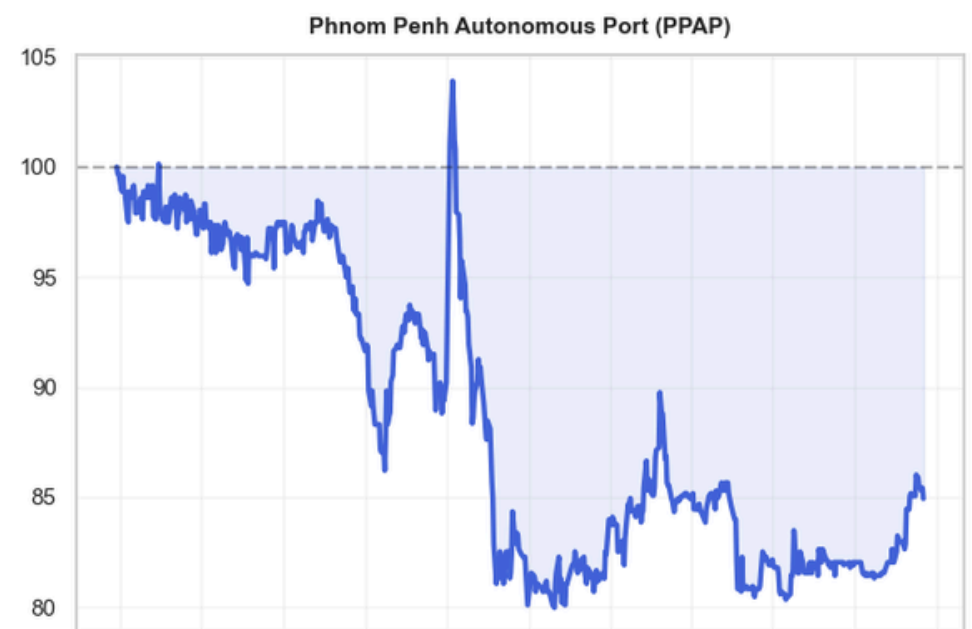
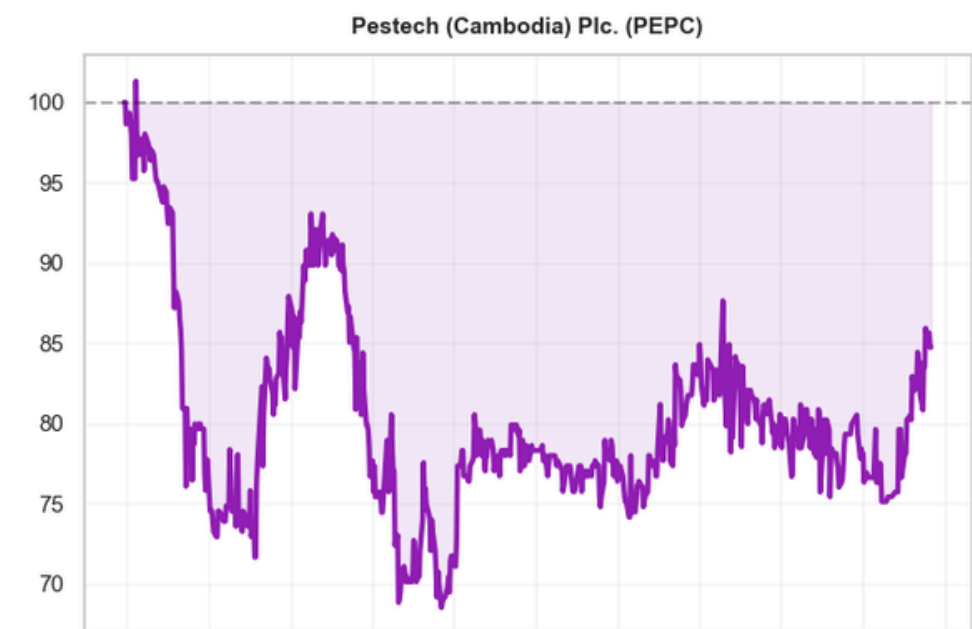
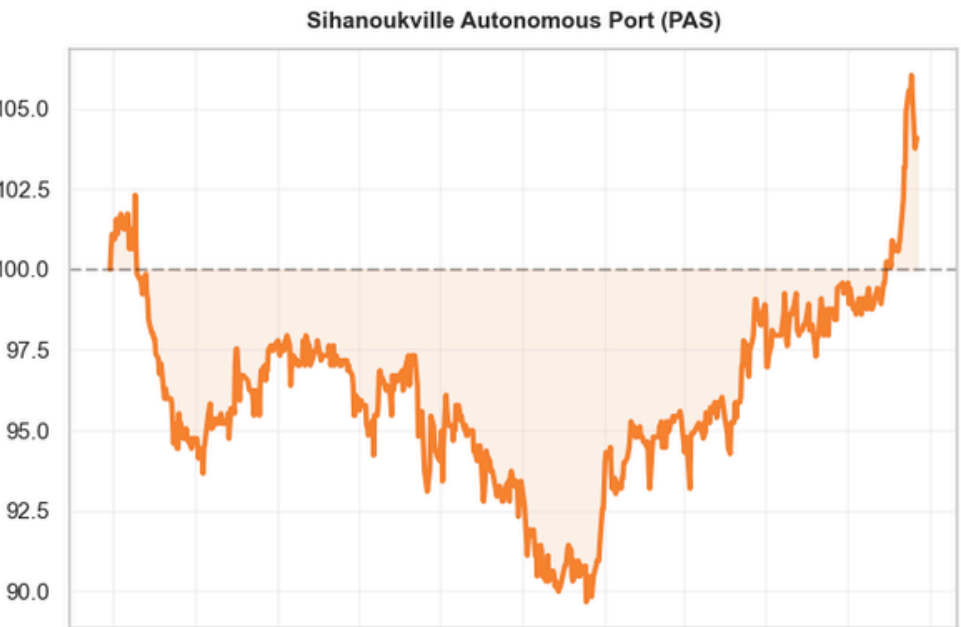
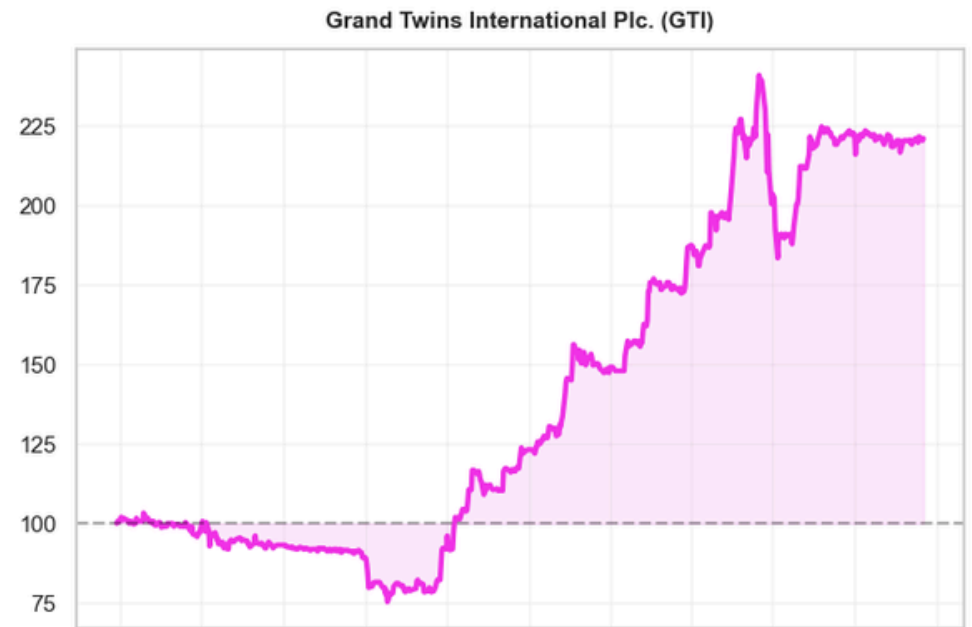
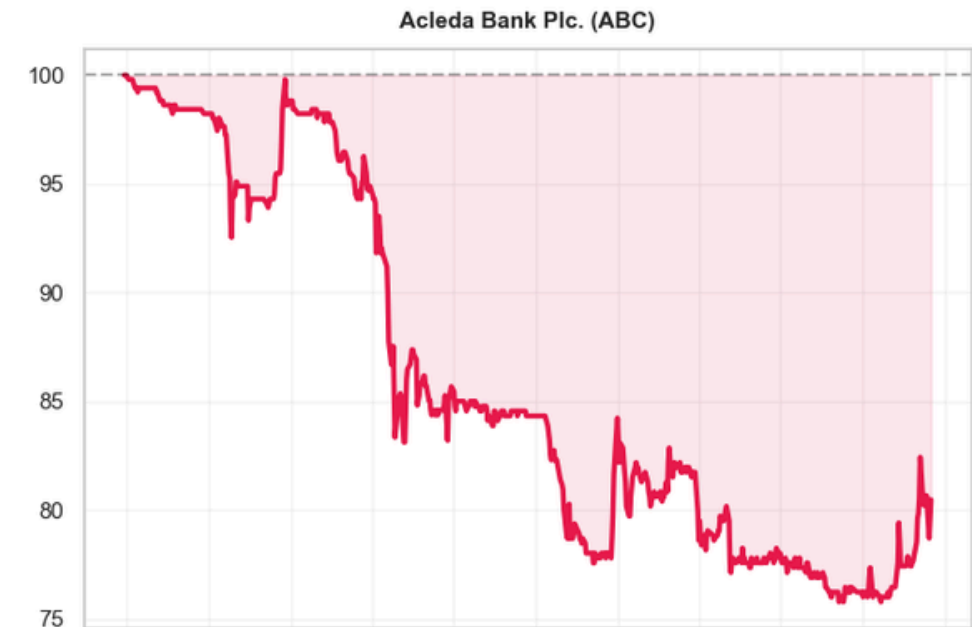
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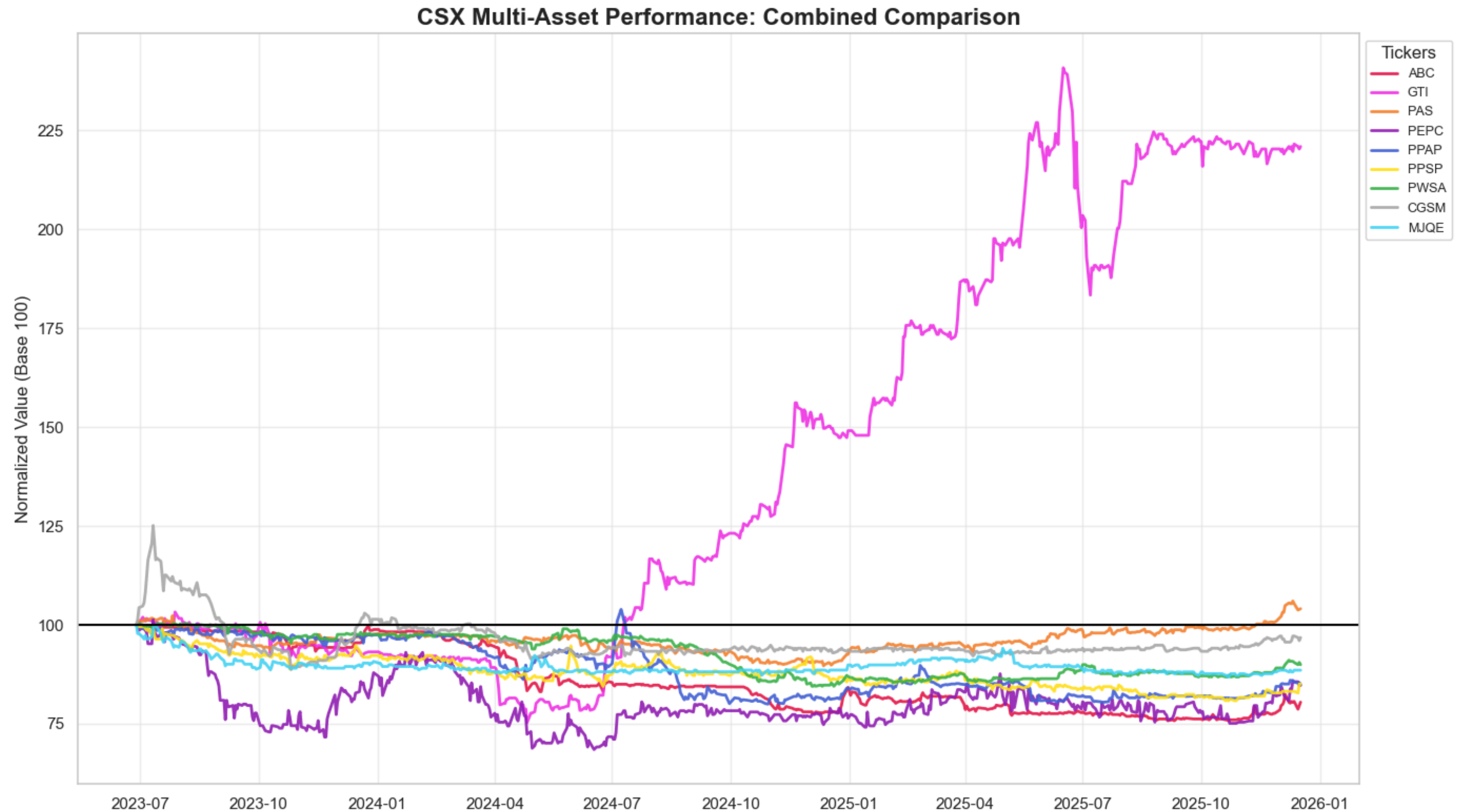
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A

Individual Rebased Price Performance (Base 100)

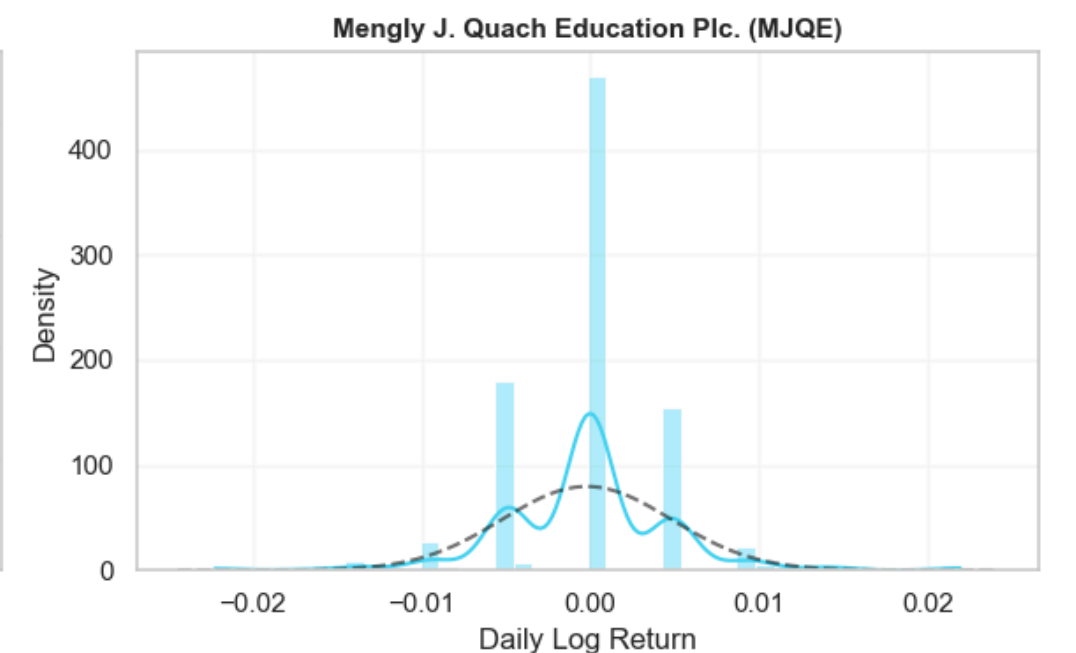
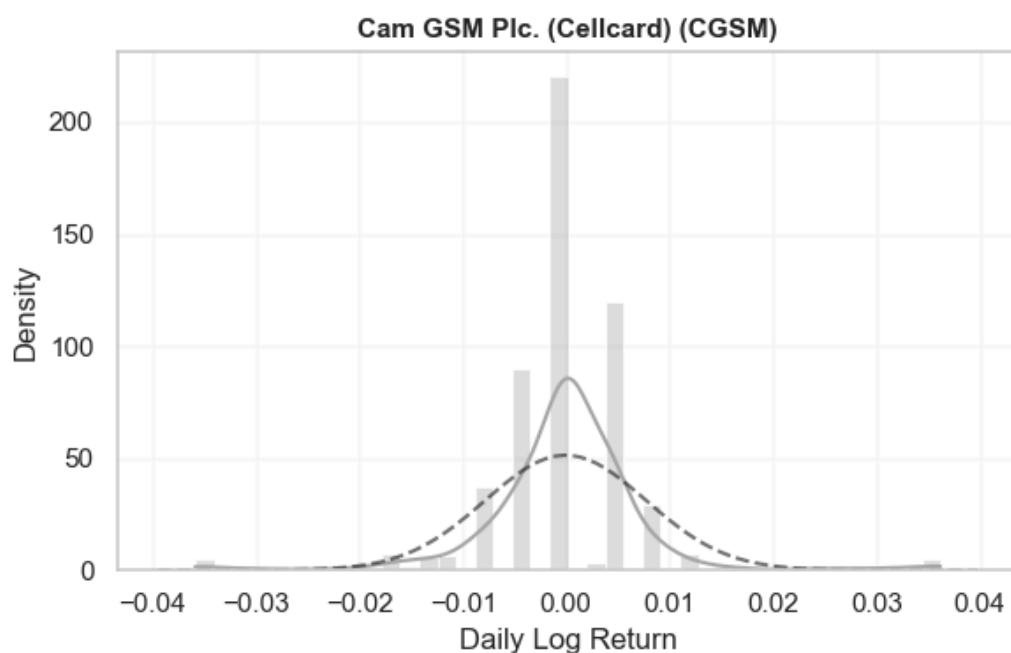
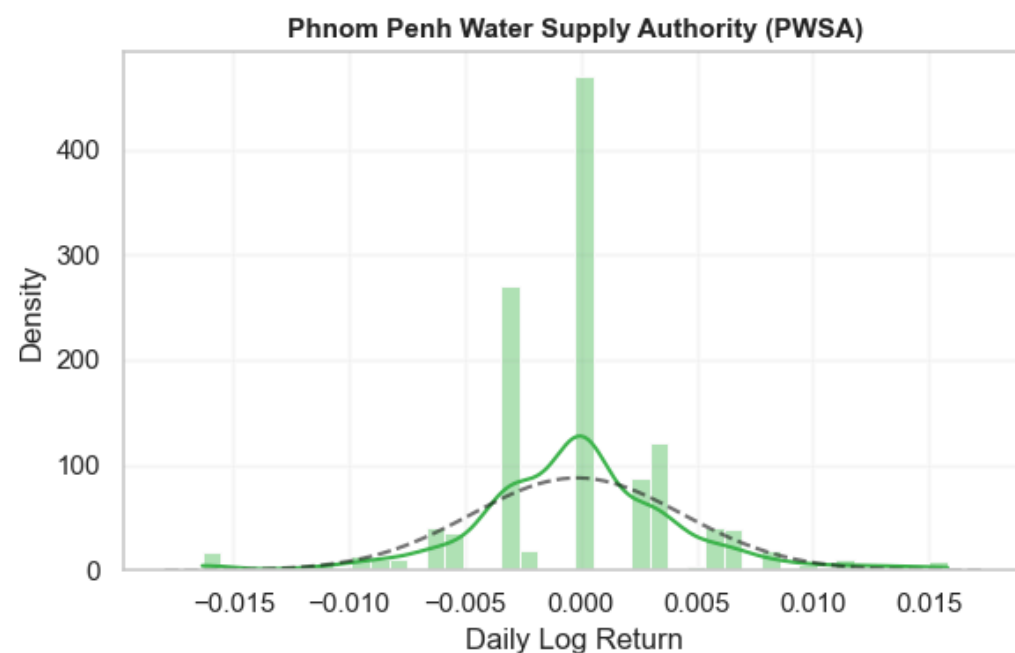
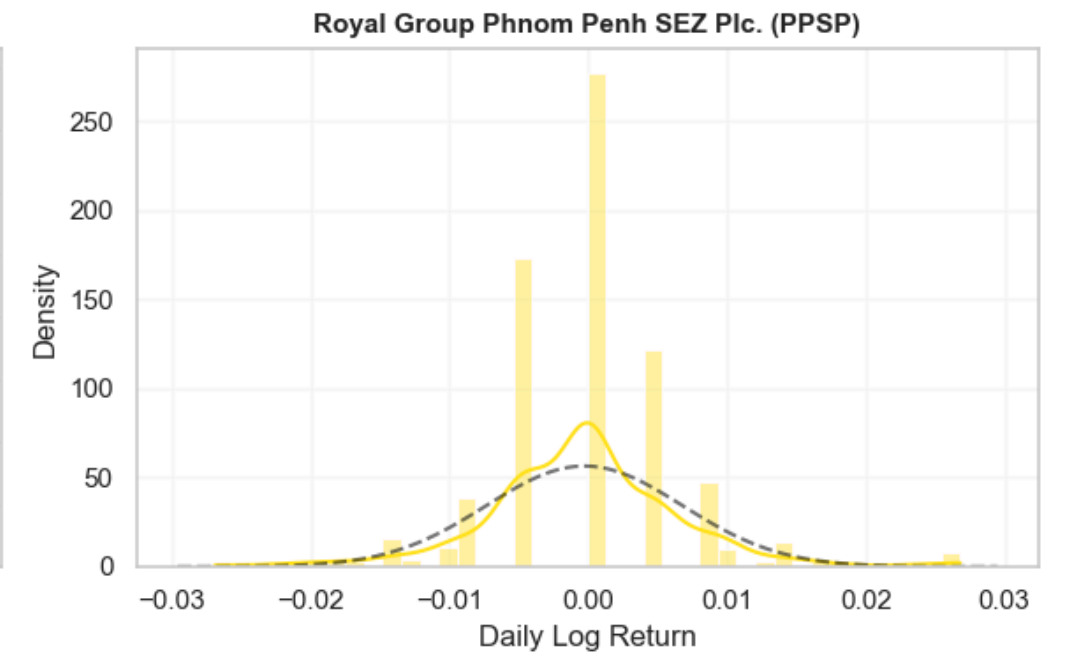
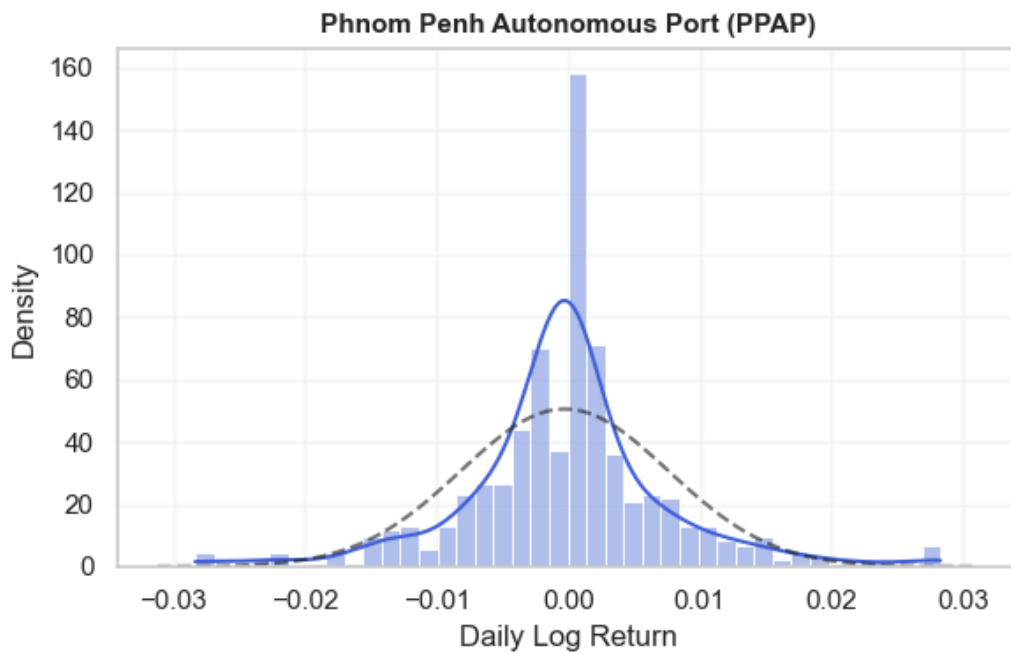
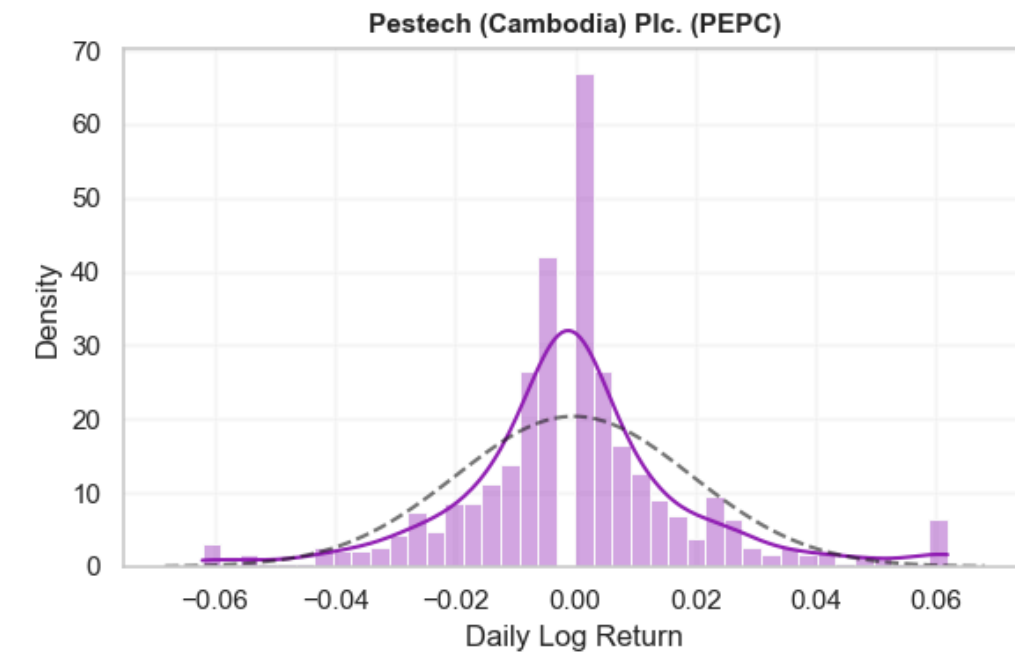
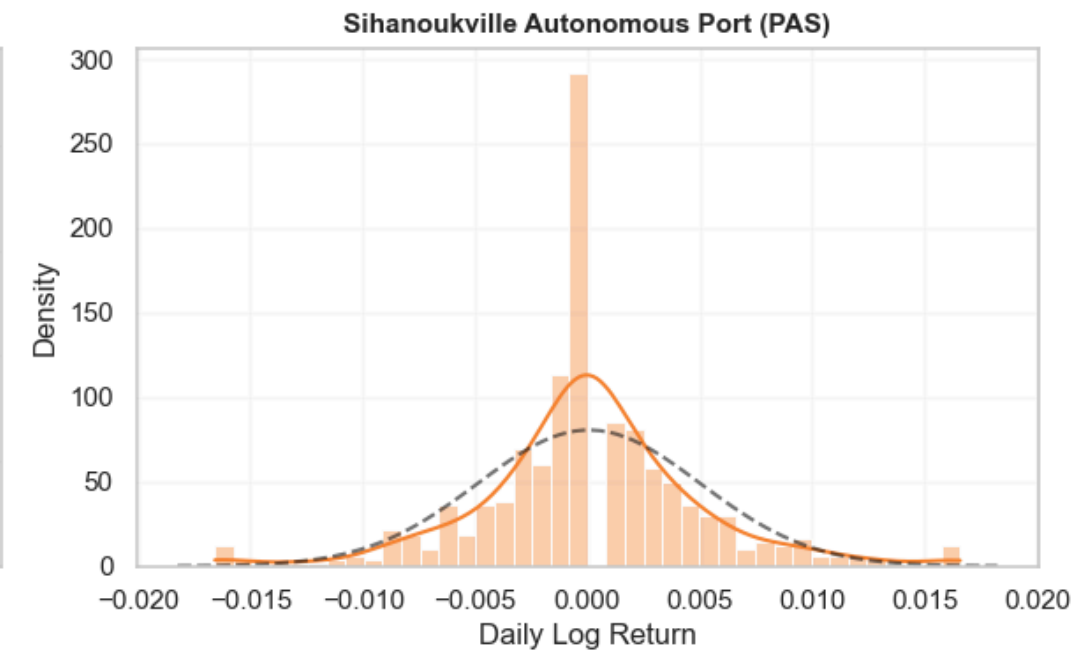
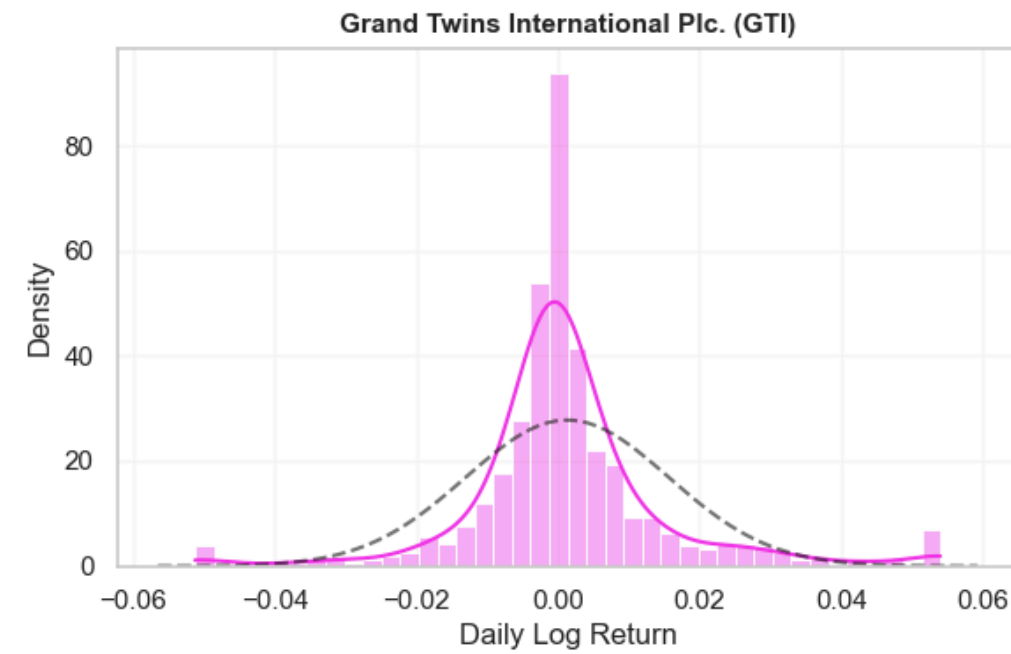
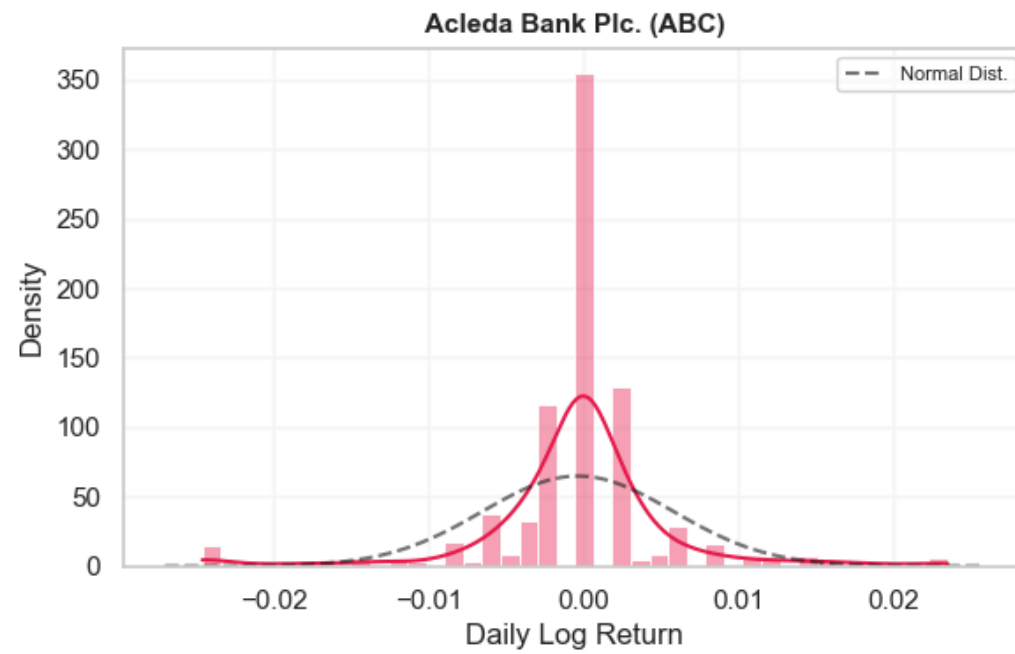


3.4 EDA



Returns Analysis: Distribution of Daily Returns vs. Normal Curve

3.4



3.4

Volatility Analysis

Risk ranking for

PWSA 0.0722

PAS 0.0787

MJQE 0.0796

ABC 0.0977

PPSP 0.1121

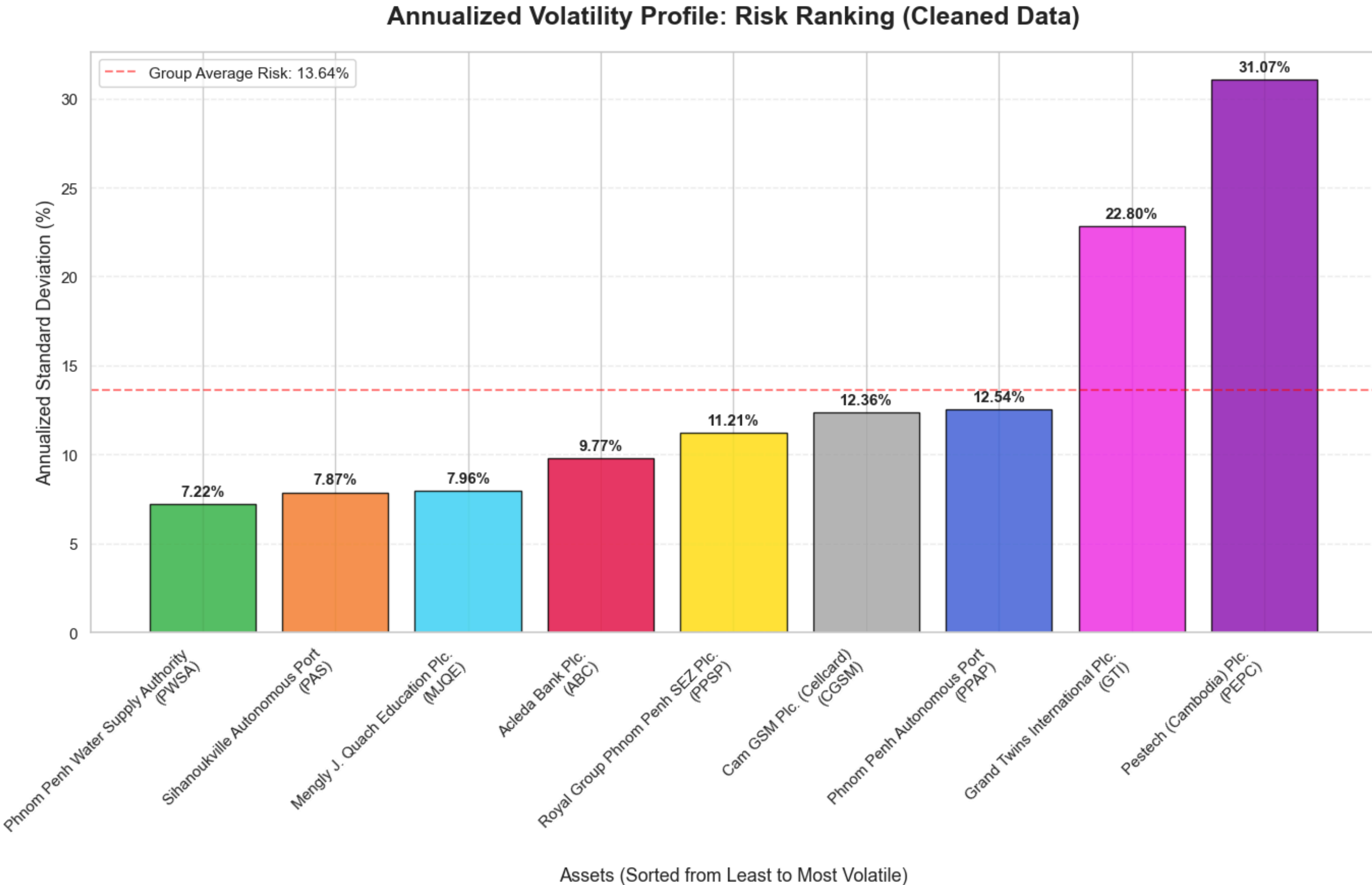
CGSM 0.1236

PPAP 0.1254

GTI 0.2280

PEPC 0.3107

dtype: float64



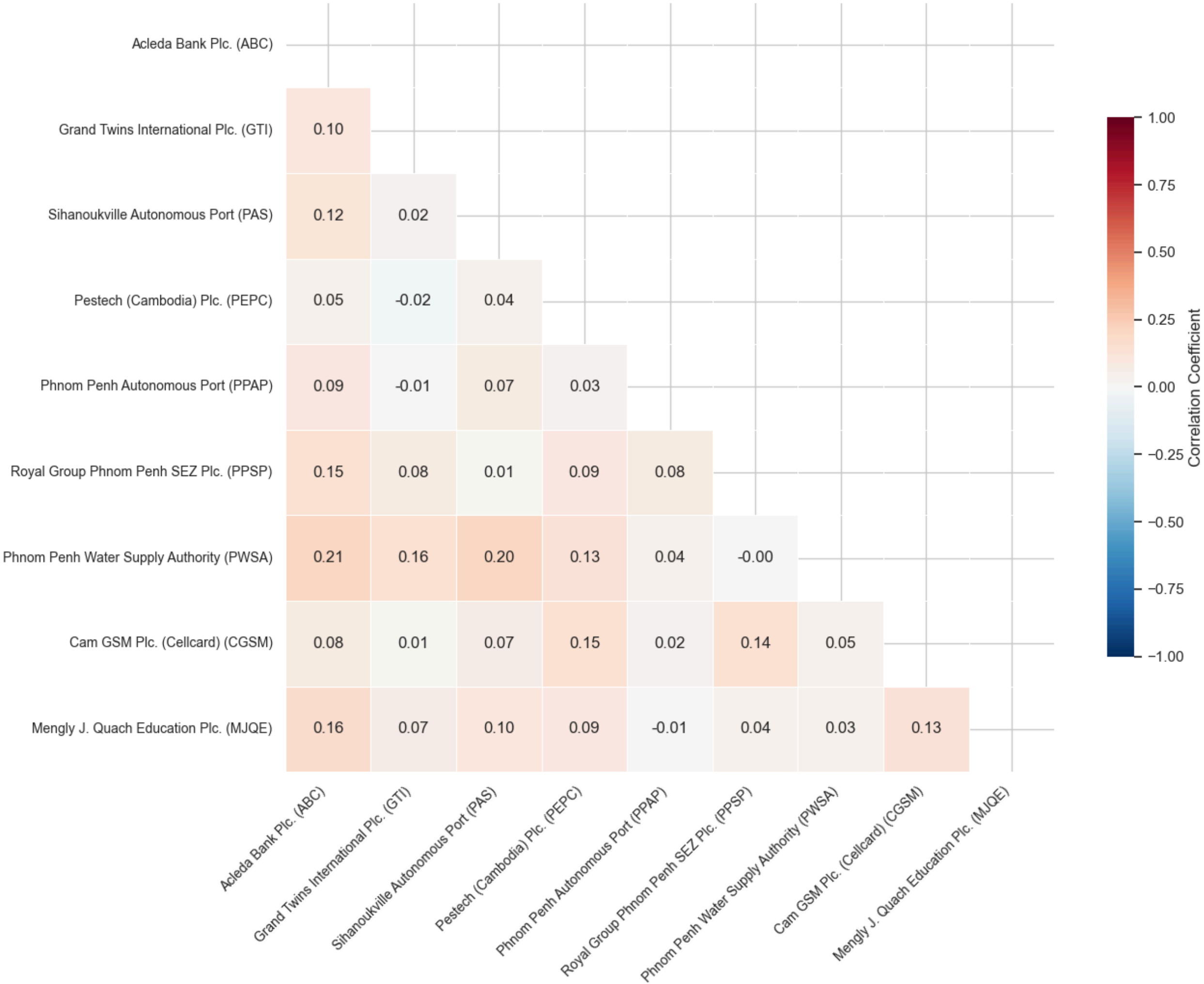
3.4

Correlation Matrix

Top 5 Diversification Opportunities (Lowest Correlations)

PEPC	GTI	-0.0167
PPAP	GTI	-0.0112
	MJQE	-0.0072
PPSP	PWSA	-0.0048
CGSM	GTI	0.0112
dtype: float64		

Asset Correlation Matrix: The Diversification Map



3.4 Covariance Matrix

	ABC	GTI	PAS	PEPC	PPAP	PPSP	PWSA	CGSM	MJQE
ABC	0.009543	0.002200	0.000902	0.001591	0.001066	0.001602	0.001485	0.001009	0.001228
GTI	0.002200	0.052007	0.000306	-0.001180	-0.000320	0.001990	0.002572	0.000315	0.001203
PAS	0.000902	0.000306	0.006190	0.001029	0.000726	0.000121	0.001128	0.000677	0.000599
PEPC	0.001591	-0.001180	0.001029	0.096540	0.000981	0.003236	0.002908	0.005604	0.002292
PPAP	0.001066	-0.000320	0.000726	0.000981	0.015714	0.001151	0.000369	0.000324	-0.000072
PPSP	0.001602	0.001990	0.000121	0.003236	0.001151	0.012563	-0.000039	0.001901	0.000346
PWSA	0.001485	0.002572	0.001128	0.002908	0.000369	-0.000039	0.005220	0.000484	0.000183
CGSM	0.001009	0.000315	0.000677	0.005604	0.000324	0.001901	0.000484	0.015270	0.001273
MJQE	0.001228	0.001203	0.000599	0.002292	-0.000072	0.000346	0.000183	0.001273	0.006331

METHODOLOGY

Understand the key stages

1



Collect data

2



Calculate
returns

3



Estimate μ
and Σ

4



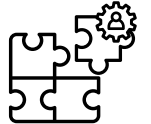
Solve
optimization
problem

5



Construct
efficient
frontier

6



Identify GMV
and optimal
portfolios

4.Implementation & Results

4.1 Case 1: WITH Short-Selling Allowed

- **4.1.1 Analytical Approach (Closed-Form Solution)**
 - **Mathematical Derivation of Closed-Form Portfolio Weights**
 - **Efficient Frontier Curve (Analytical)**
- **4.1.2 Validation: Analytical vs. Numerical**
 - **Comparison of GMV Weights (Analytical vs. Numerical)**
 - **Compare Both Efficient Frontiers**
 - **Discussion: Agreement validates correct formula implementation**

4.2 Case 2: WITHOUT Short-Selling (Long-Only Constraint)

4.3 Comparative Analysis Across All Cases

4.Implementation & Results

4.1 Case 1: WITH Short-Selling Allowed

4.1.1 Analytical Approach (Closed-Form Solution)

Global Minimum Variance

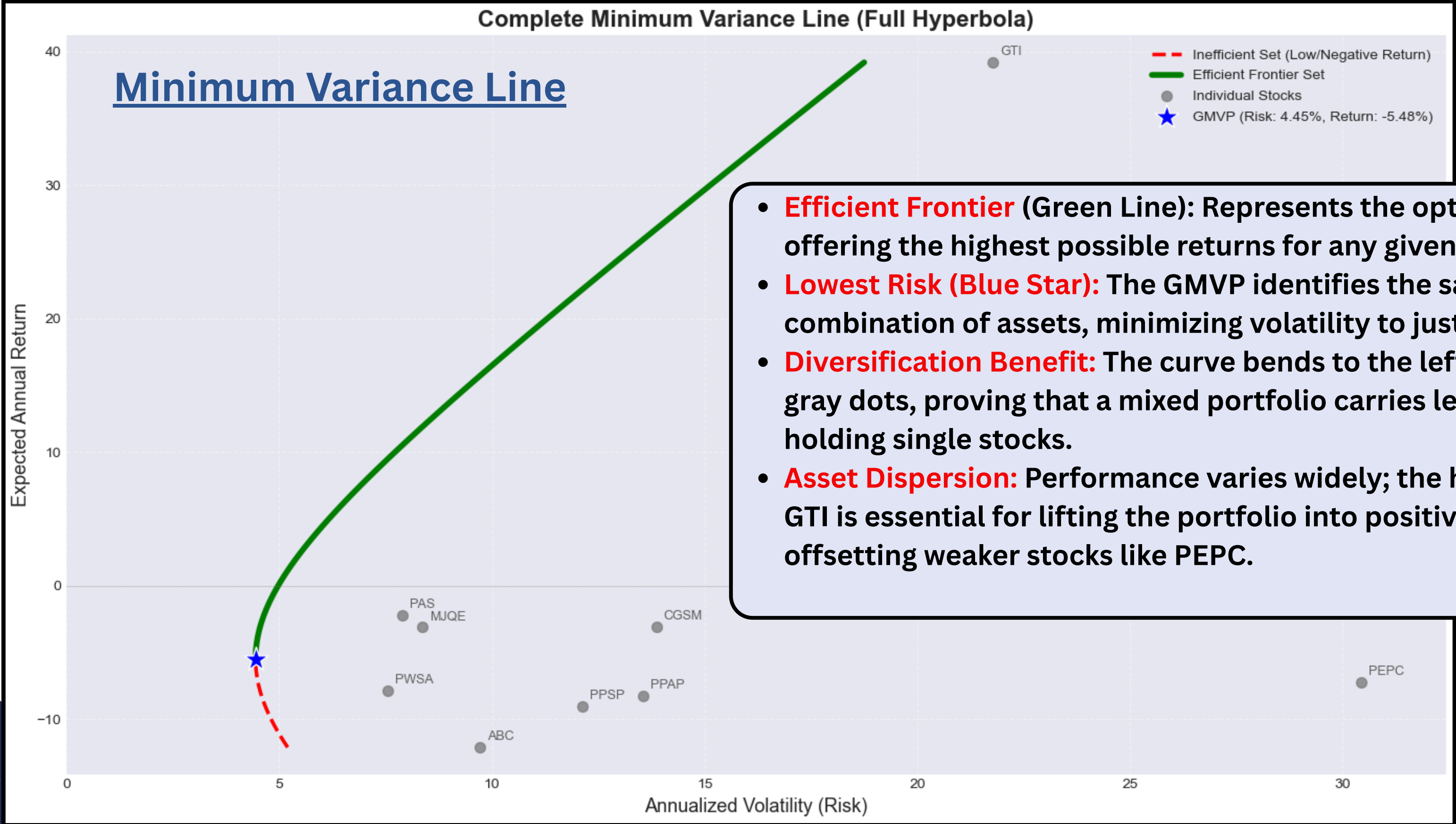
Global Minimum Variance Portfolio Weights (Short Selling Allowed)

Ticker	Asset Name	Weight (%)
PWSA	Phnom Penh Water Supply Authority	26.23
MJQE	Mengly J. Quach Education Plc.	22.12
PAS	Sihanoukville Autonomous Port	21.82
PPSP	Royal Group Phnom Penh SEZ Plc.	10.32
PPAP	Phnom Penh Autonomous Port	7.43
ABC	ACLEDA Bank Plc.	7.19
CGSM	Cambodia GSM Plc. (Cellcard)	4.53
GTI	Grand Twins International Plc.	0.70
PEPC	Pestech (Cambodia) Plc.	−0.35

Expected Return: −5.4783%
Volatility: 4.4461%

4.Implementation & Results

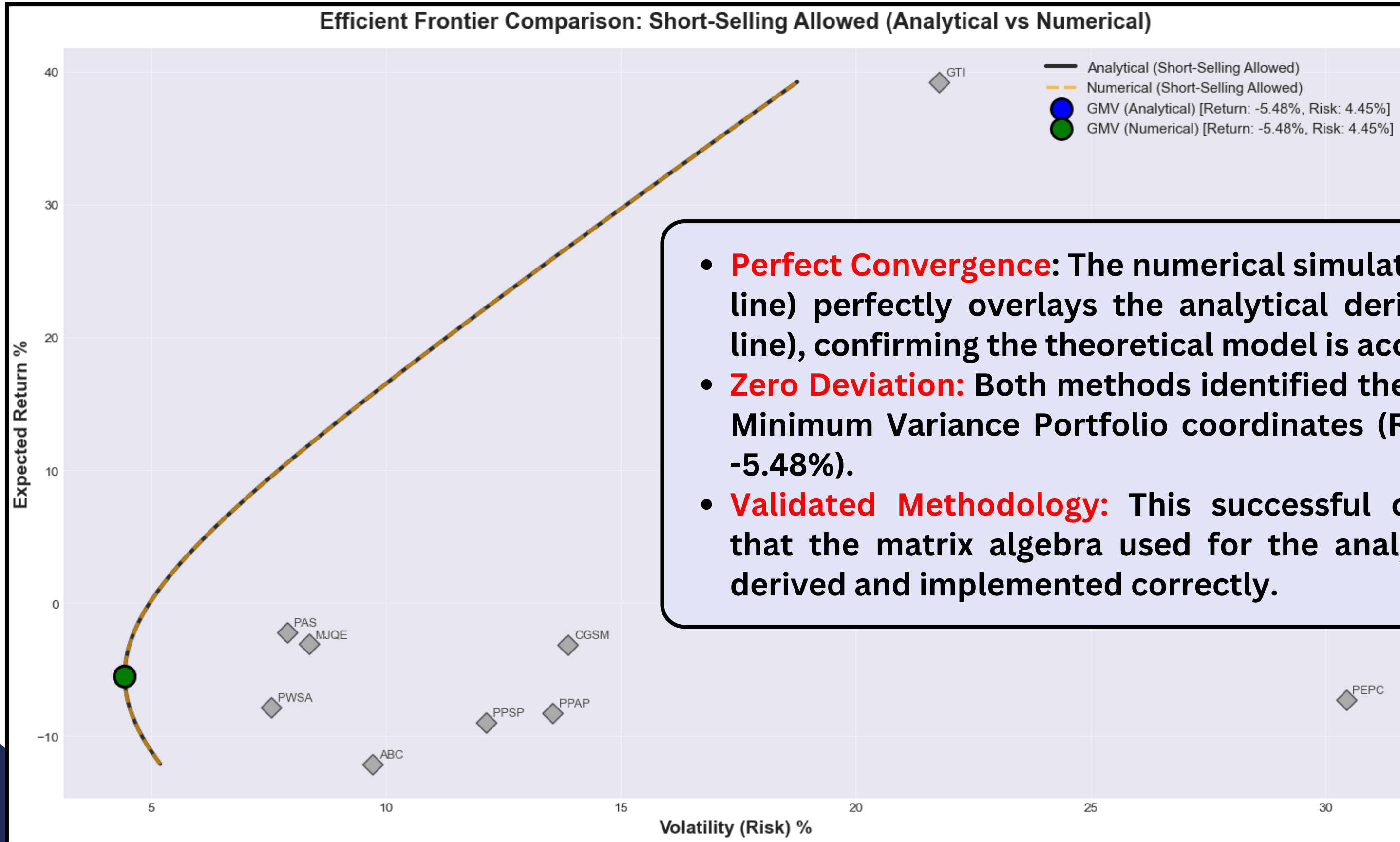
4.1.1 Analytical Approach (Closed-Form Solution)



- **Efficient Frontier** (Green Line): Represents the optimal portfolios offering the highest possible returns for any given risk level.
- **Lowest Risk (Blue Star)**: The GMVP identifies the safest possible combination of assets, minimizing volatility to just **4.45%**.
- **Diversification Benefit**: The curve bends to the left of the individual gray dots, proving that a mixed portfolio carries less risk than holding single stocks.
- **Asset Dispersion**: Performance varies widely; the high-return outlier GTI is essential for lifting the portfolio into positive territory, offsetting weaker stocks like PEPC.

4.Implementation & Results

4.1.2 Validation: Analytical vs. Numerical



- **Perfect Convergence:** The numerical simulation (dashed orange line) perfectly overlays the analytical derivation (solid black line), confirming the theoretical model is accurate.
- **Zero Deviation:** Both methods identified the exact same Global Minimum Variance Portfolio coordinates (Risk: 4.45%, Return: -5.48%).
- **Validated Methodology:** This successful cross-check proves that the matrix algebra used for the analytical solution was derived and implemented correctly.

4.Implementation & Results

4.2 WITHOUT Short-Selling (Long-Only Constraint)

Global Minimum Variance

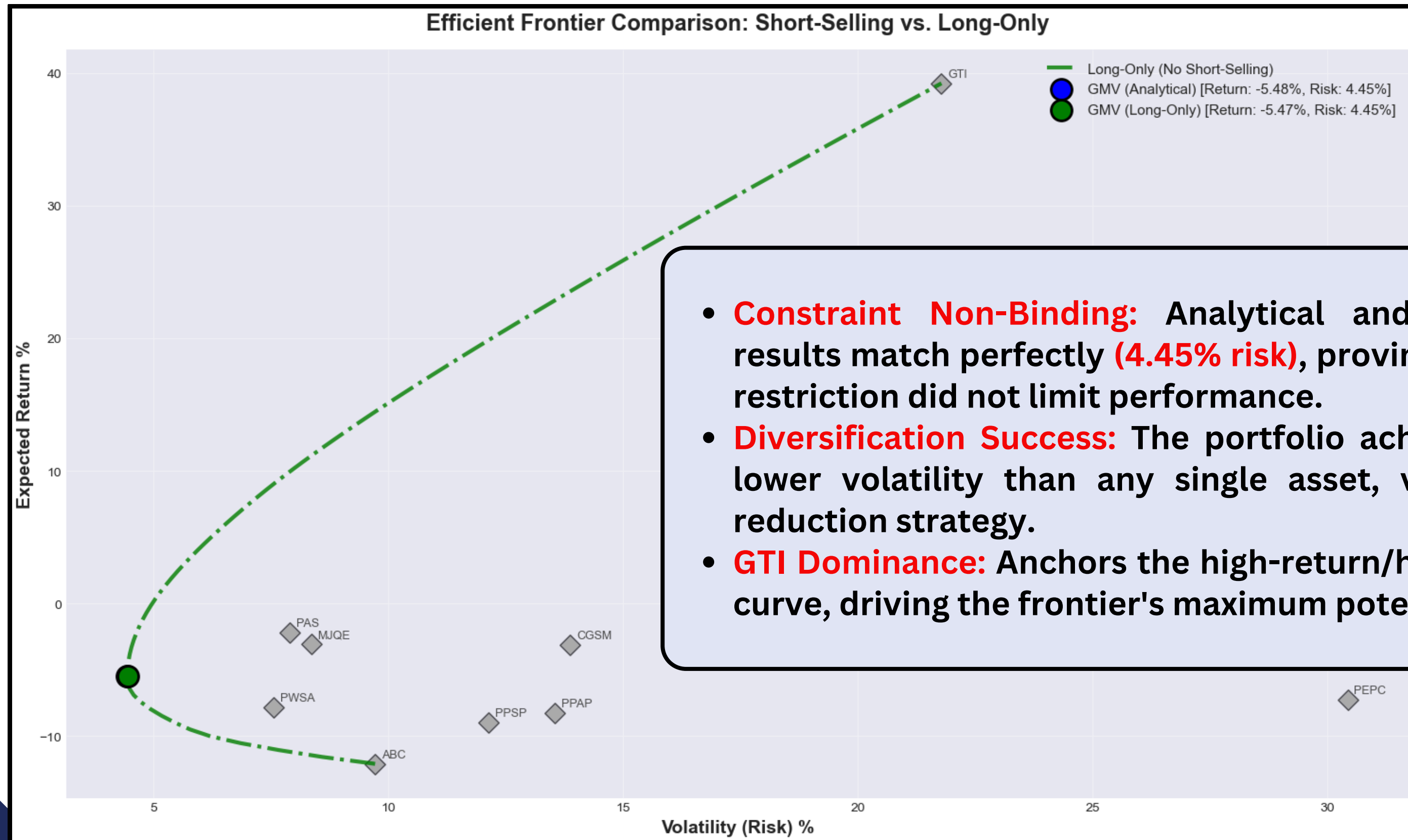
Table 10: Global Minimum Variance Portfolio Weights (No Short Selling)

Ticker	Weight (%)
PWSA	26.08
MJQE	22.08
PAS	21.80
PPSP	10.21
ABC	7.23
PPAP	7.42
CGSM	4.45
GTI	0.72
PEPC	0.00

Expected Return: -5.4725%
Volatility: 4.4473%

4.Implementation & Results

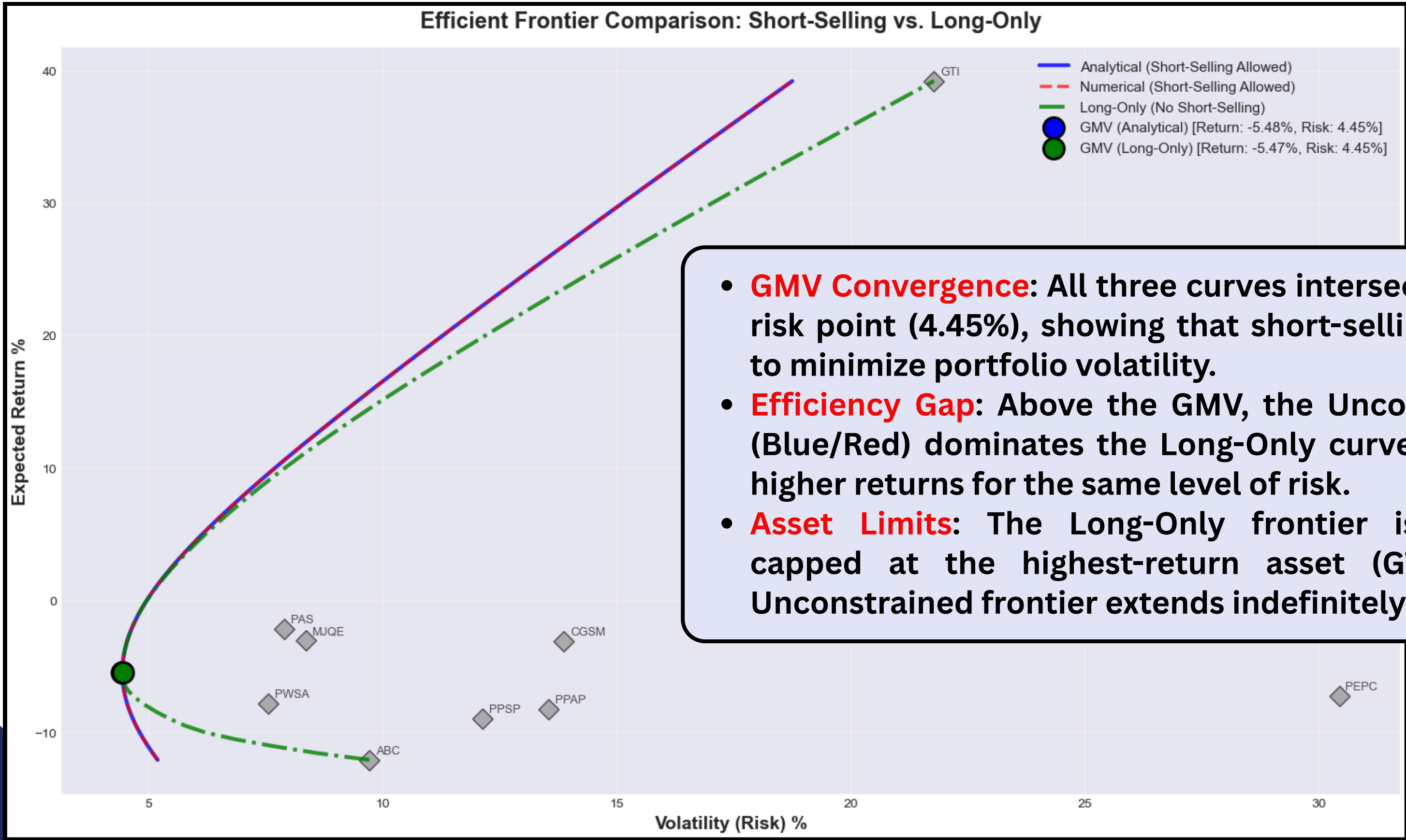
4.2 WITHOUT Short-Selling (Long-Only Constraint)



- **Constraint Non-Binding:** Analytical and Long-Only GMV results match perfectly (**4.45% risk**), proving the short-selling restriction did not limit performance.
- **Diversification Success:** The portfolio achieves significantly lower volatility than any single asset, validating the risk reduction strategy.
- **GTI Dominance:** Anchors the high-return/high-risk end of the curve, driving the frontier's maximum potential.

4.Implementation & Results

4.3 Comparative Analysis Across All Cases



- **GMV Convergence:** All three curves intersect at the minimum risk point (4.45%), showing that short-selling is not required to minimize portfolio volatility.
- **Efficiency Gap:** Above the GMV, the Unconstrained frontier (Blue/Red) dominates the Long-Only curve (Green), offering higher returns for the same level of risk.
- **Asset Limits:** The Long-Only frontier is mathematically capped at the highest-return asset (GTI), whereas the Unconstrained frontier extends indefinitely via leverage.

4.Implementation & Results

4.3 Comparative Analysis Across All Cases

=====
SHORT-SELLING IMPACT ANALYSIS
=====
1. GLOBAL MINIMUM VARIANCE (GMV) PORTFOLIO

WITH Short-Selling:
Return: -5.4771%, Volatility: 4.4461%
Negative Weights: 1 assets
 WITHOUT Short-Selling (Long-Only):
Return: -5.4725%, Volatility: 4.4473%
Zero Weights: 1 assets
 Impact:
Δ Return: +0.0046%
Δ Volatility: +0.0012%

Key Findings:

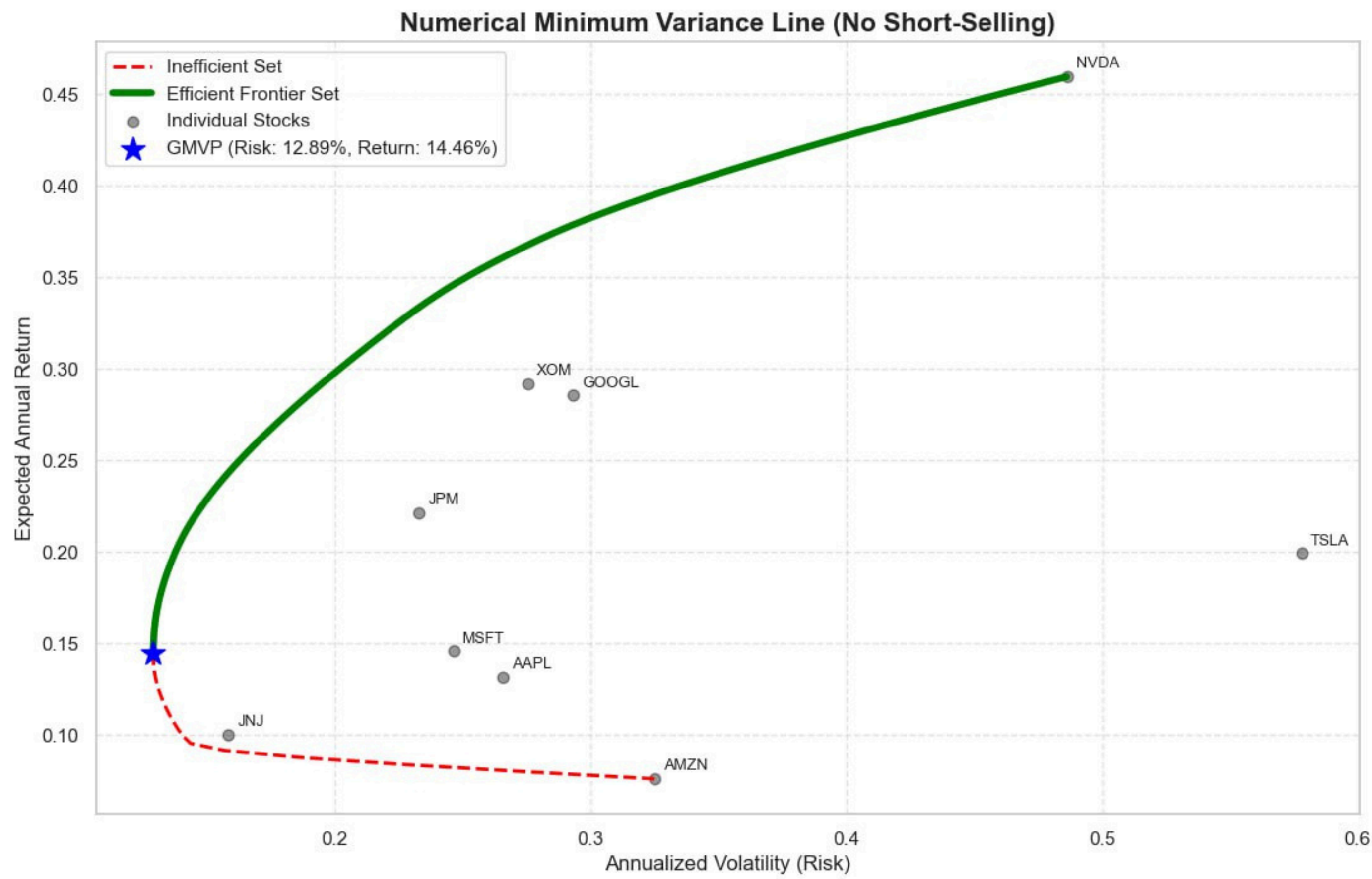
1. Efficient Frontier Shape: The long-only frontier lies to the right of (is dominated by) the unconstrained frontier, indicating that short-selling restrictions reduce portfolio efficiency.

2. Risk-Return Trade-off: Long-only portfolios typically have:

- Higher volatility for the same expected return, OR
- Lower expected return for the same volatility

3. Portfolio Concentration: Long-only constraints often lead to more concentrated portfolios with some assets receiving zero weight.

4.Implementation & Results



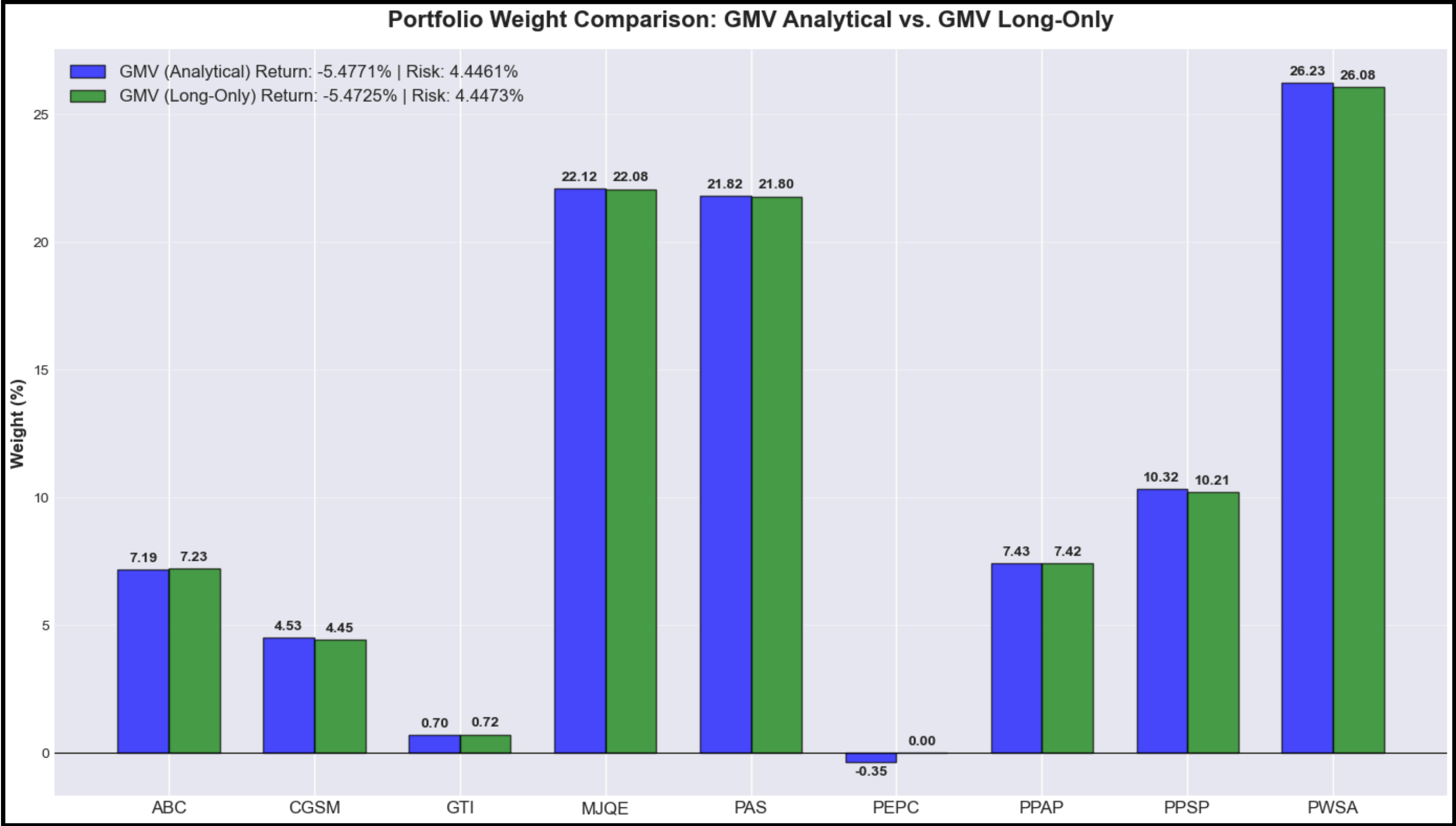
4.Implementation & Results

4.4 Portfolio Performance Metrics

=====		
IN-SAMPLE PERFORMANCE METRICS		
=====		
Portfolio	Expected Return (%)	Volatility (%)
GMV (Analytical)	-5.477111	4.446099
GMV (Numerical)	-5.478251	4.446100
GMV (Long-Only)	-5.472544	4.447290
=====		

4.Implementation & Results

4.5 Portfolio Portfolio Weights Visualization



4. Implementation & Results

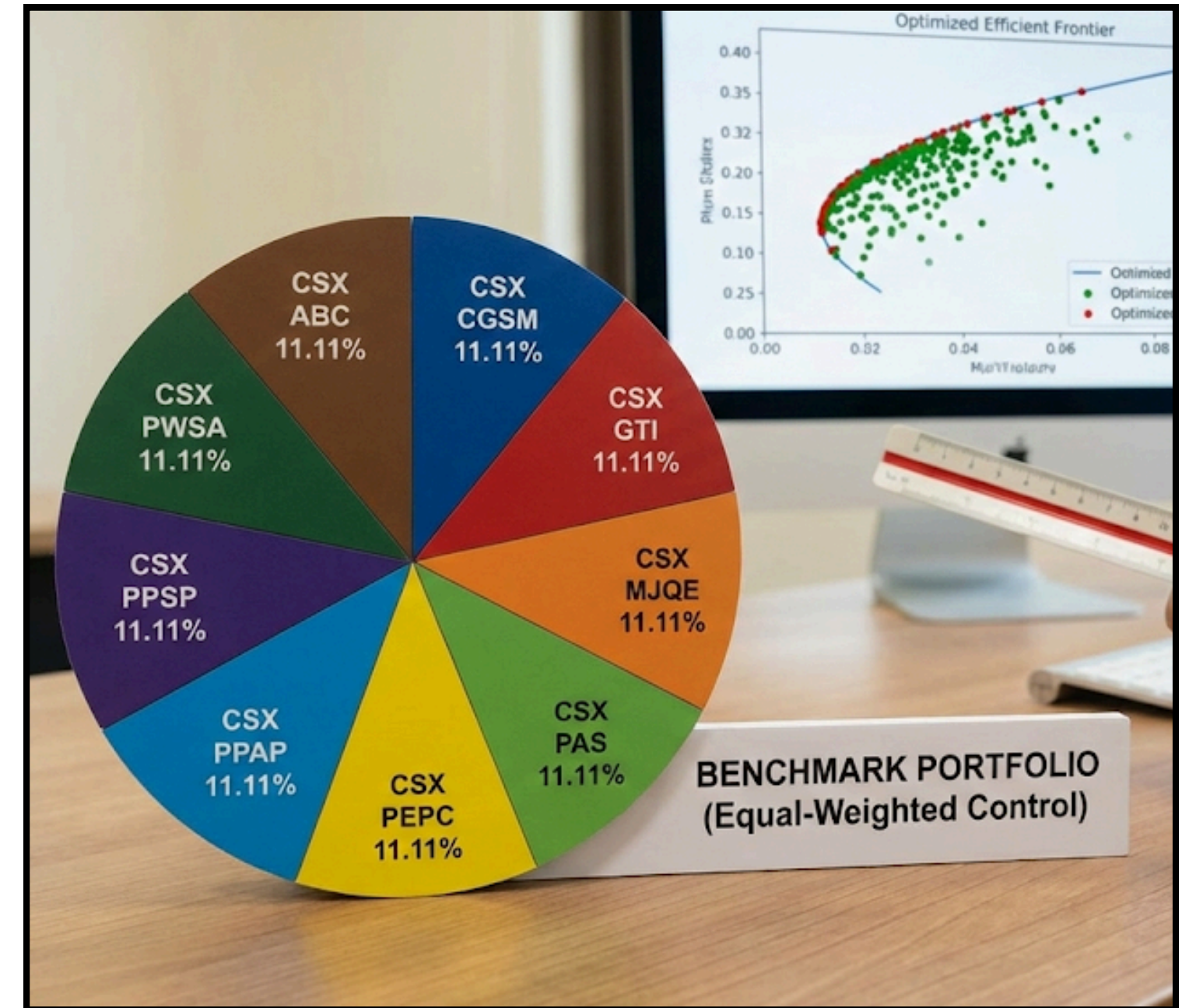
4.6 Out-of-Sample Backtesting & Performance Validation

4.6.1 Benchmark Portfolio Construction (Equal-Weighted)

Concept: The Benchmark (The Control Group)

Why do we need this? Before we test our "smart" optimized portfolios, we need a standard to beat. In finance, we often use an **Equal-Weighted** Portfolio as a baseline.

- The Logic: If we simply put 11.11% (1/9th) of our money into every CSX stock, we get a diversified but "unintelligent" portfolio.
- The Test: If our complex GMV model cannot beat this simple strategy (i.e., if it has higher risk), then our **optimization failed**.



4. Implementation & Results

4.6 Out-of-Sample Backtesting & Performance Validation

4.6.1 Benchmark Portfolio Construction (Equal-Weighted)

Equal-Weighted Benchmark Portfolio:

Weight per asset: 11.11%

In-Sample Metrics:

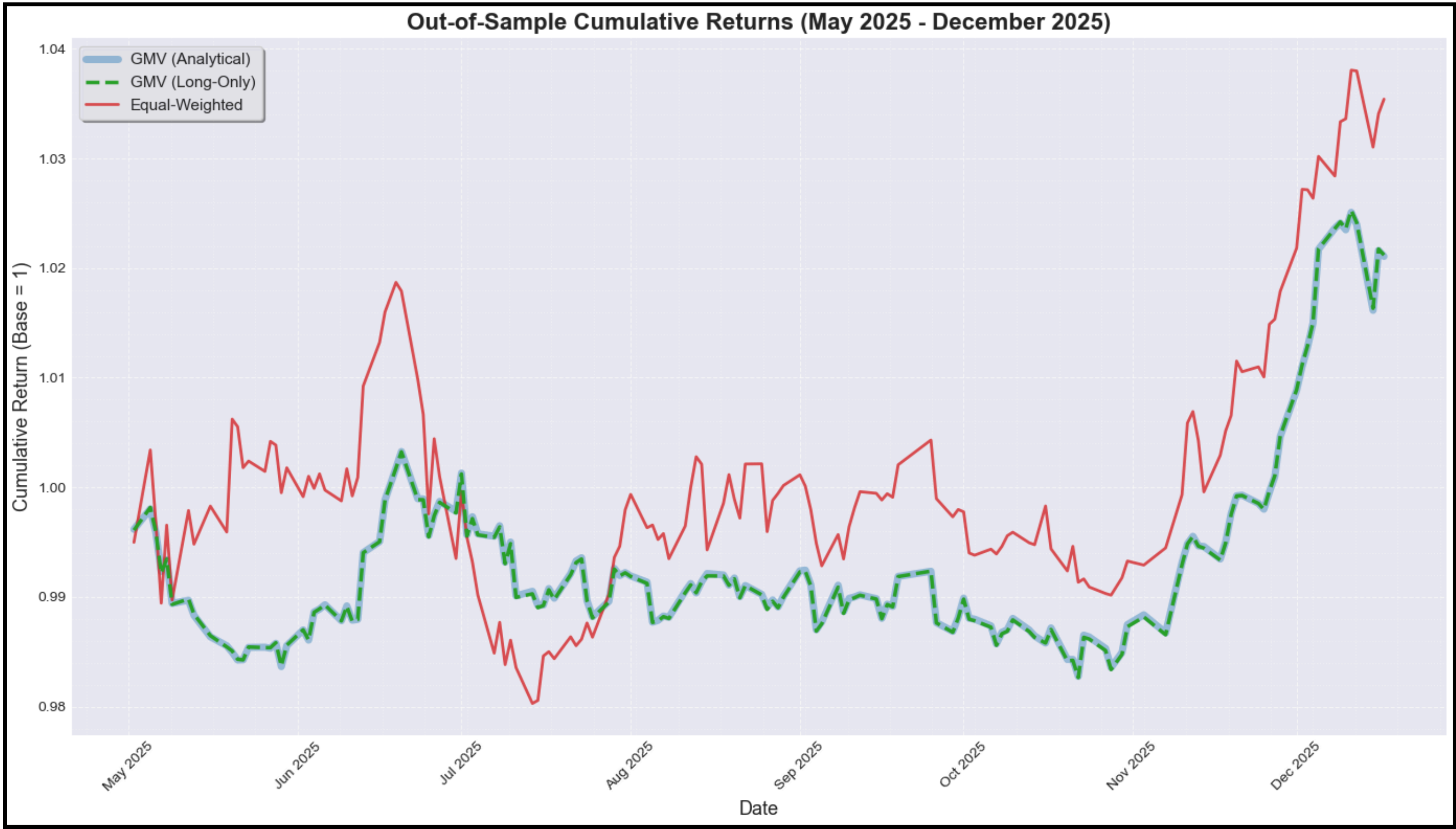
Expected Return: -1.5159%

Volatility: 6.2962%

- Did we beat the Benchmark? YES.
- The GMV portfolio risk is (4.45%)% which is lower than the Benchmark risk (6.29%).

4.Implementation & Results

4.6.2 Out-of-Sample Performance Results



4.Implementation & Results

4.6.2 Out-of-Sample Performance Results

=====							
EXPECTED VS. REALIZED PERFORMANCE							
=====							
Portfolio	Expected Return (%)	Realized Return (%)	Expected Vol (%)	Realized Vol (%)	Return Gap (%)	Vol Gap (%)	
GMV (Analytical)	-5.477111	3.493968	4.446099	3.482523	8.971079	-0.963576	
GMV (Long-Only)	-5.472544	3.514465	4.447290	3.493379	8.987010	-0.953910	
Equal-Weighted	-1.515912	5.884579	6.296216	5.548398	7.400491	-0.747818	
=====							

- **Returns Defied Forecasts:** Realized gains 3.49% to 5.88% outperformed negative predictions.
- **Risk Overestimated:** Realized volatility was 0.75%to 0.96% lower than expected.
- **GMV Efficacy:** Minimized risk 3.48% vol but trailed the benchmark in returns.
- **Equal-Weighted:** Delivered the highest return 5.88% but with the highest volatility 5.55%.
- **Constraints:** Long-Only constraint had negligible impact on performance.

Key Takeaway:

- Market conditions were significantly more favorable and stable than the historical models predicted.

5.Sensitivity Analysis

5.1 Impact of Correlation Changes on Efficient Frontier Shape

5.2 How Individual Asset Volatility Affects the Efficient Frontier

5.3 Effect of Expected Return Assumptions

(Check Our Report for this Senisitive Analysis Section)

6. Conclusions & Recommendations

6.1. Executive Summary of Findings

What we did:

- Applied Markowitz model to 9 CSX stocks.
- Evaluated the gap between theoretical optimality and real-world stability.

Key Findings:

- **Risk Reduction Works:** The GMV portfolio reduced risk compared to the market benchmark, showing that diversification works even in the CSX.
- **Short-Selling (Theory vs. Practice):** Short-selling improves portfolio in theory, but in practice it is costly and difficult in the CSX due to low liquidity.
- **Out-of-Sample Reality:** Optimal portfolios did not consistently outperform a simple equal-weighted strategy, mainly because future returns are harder to predict.

6. Conclusions & Recommendations

6.2. Math vs. Reality

Theory

- Maximizes Sharpe ratio
- Uses expected returns
- Mathematically optimal

Reality

- Highly unstable weights
- Returns hard to estimate
- Weak out-of-sample results

Key takeaways:

- GMV portfolios (covariance-only) proved most robust.
- Simpler models often perform better in practice.

6.Conclusions & Recommendations

6.3. Investment Recommendations

Investor Profile	Recommended Strategy	Evidence / Rationale
Conservative / Stability-Focused	Global Minimum Variance (Long-Only)	Why: Achieved the lowest realized volatility in our backtests. Strength: Its weights depend only on covariance (risk), which is more stable and predictable than returns. It avoids the "Estimation Trap."
Aggressive / Alpha-Seeking	Optimal Portfolio (Mean-Variance)	Why: Offers the highest <i>theoretical</i> efficiency. Condition: Only recommended if you have high confidence in your return forecasts (e.g., proprietary fundamental models). Tip: Combine with a 50% GMV "anchor" to dampen volatility.
Passive / Low-Maintenance	Equal Weighted	Why: Surprisingly competitive risk-adjusted returns (Sharpe Ratio) in our out-of-sample tests. Strength: Zero optimization complexity, lower turnover costs, and immunity to model error.

6. Conclusions & Recommendations

6.4. Limitations & Future Work

Limitation

- CSX Market is limited and liquidity.
- Data quality and availability (MJQE, CGSM)
- Model assumptions (normality, constant parameters.
- Backtesting limitations (short history, single test period)

Future Work

- **Robust estimation methods:** Consider shrinkage estimations or Bayesian methods for more stable estimates.
- **Dynamic strategies:** Multi-period optimization with rebalancing.
- **Incorporate transaction costs**
- **Expand to regional diversification**

6. Conclusions & Recommendations

Key Takeaways:

- **Diversification Works:** Even in small markets, risk reduction achievable.
- **Simplicity is Better:** Covariance-based GMV more stable than return-dependent optimal.
- **Theory is not the same as Practice:** Estimation error matters more than theoretical efficiency.
- **Know Your Goals:**
 - Conservative investors → GMV
 - Aggressive → Rebalanced optimal

7. Appendix & Q&A

Full Covariance Matrix

```
.. print("Full Annualized Covariance Matrix (In-Sample):")
   print("=" * 100)
   print(Sigma_train)
   print("=" * 100)
```

Full Annualized Covariance Matrix (In-Sample):

```
=====
```

	ABC	CGSM	GTI	MJQE	PAS	PEPC	PPAP	\
ABC	0.009451	0.001249	0.003166	0.001582	0.000732	0.000682	0.001241	
CGSM	0.001249	0.019235	0.000902	0.001576	0.000898	0.006421	0.000470	
GTI	0.003166	0.000902	0.047425	0.001753	0.000240	-0.000414	-0.000206	
MJQE	0.001582	0.001576	0.001753	0.006984	0.000642	0.001597	0.000154	
PAS	0.000732	0.000898	0.000240	0.000642	0.006238	0.001435	0.000723	
PEPC	0.000682	0.006421	-0.000414	0.001597	0.001435	0.092564	0.001502	
PPAP	0.001241	0.000470	-0.000206	0.000154	0.000723	0.001502	0.018354	
PPSP	0.001792	0.002699	0.003301	0.000420	0.000092	0.004757	0.001849	
PWSA	0.001656	0.000660	0.002320	0.000172	0.001220	0.002637	0.000483	

	PPSP	PWSA
ABC	0.001792	0.001656
CGSM	0.002699	0.000660
GTI	0.003301	0.002320
MJQE	0.000420	0.000172
PAS	0.000092	0.001220
PEPC	0.004757	0.002637
PPAP	0.001849	0.000483
PPSP	0.014706	-0.000189
PWSA	-0.000189	0.005718

```
=====
```

7. Appendix & Q&A

Complete Weight Tables

```
L... print("Complete Portfolio Weights (All Methods):")
      print("=" * 100)
      print(weights_summary)
      print("=" * 100)
```

Complete Portfolio Weights (All Methods):

```
=====
  Ticker  GMV (Analytical)  GMV (Numerical)  GMV (Long-Only)
0    ABC          7.185628        7.185393        7.230014
1   CGSM          4.532804        4.528139        4.450195
2    GTI          0.701951        0.700102        0.723287
3   MJQE         22.118636       22.126452       22.082541
4    PAS         21.823094       21.819114       21.798129
5   PEPC         -0.347092       -0.347661        0.000000
6   PPAP          7.429135        7.419300        7.424040
7   PPSP         10.322786       10.343994       10.208024
8   PWSA         26.233059       26.225166       26.083772
=====
```

7.Appendix & Q&A

Detailed Performance Metrics

```
1... print("In-Sample Performance Metrics:")
    print("=" * 90)
    print(performance_summary)
    print("\n" + "=" * 90)
    print("\nOut-of-Sample Performance Metrics:")
    print("=" * 90)
    print(oos_summary)
    print("=" * 90)
```

In-Sample Performance Metrics:

=====

	Portfolio	Expected Return (%)	Volatility (%)
0	GMV (Analytical)	-5.477111	4.446099
1	GMV (Numerical)	-5.478251	4.446100
2	GMV (Long-Only)	-5.472544	4.447290

=====

Out-of-Sample Performance Metrics:

=====

	Portfolio	Realized Return (%)	Realized Volatility (%) \
0	GMV (Analytical)	3.493968	3.482523
1	GMV (Long-Only)	3.514465	3.493379
2	Equal-Weighted	5.884579	5.548398

Max Drawdown (%)

0	-2.043705
1	-2.052078
2	-3.770323

=====

7. Appendix & Q&A

References

Capiński, M., & Zastawniak, T. (2003). Mathematics for finance: An introduction to financial engineering. Springer.

Thank You

Feel free to us any questions.